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A PROJECT REPORT ON
DENTAMIND AI: AN UNCERTAINTY-AWARE PIPELINE FOR TOOTH DETECTION, FDI NUMBERING, SEGMENTATION AND FINDING CLASSIFICATION ON PANORAMIC DENTAL RADIOGRAPHS

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OF

B. TECH. IN COMPUTER ENGINEERING

FOR THE SUBJECT

GENERATIVE AI MDM CAPSTONE PROJECT
UNDER THE DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI & ML)

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2026-2027

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2026-2027

CERTIFICATE

This is to certify that the project report entitled
"DENTAMIND AI: AN UNCERTAINTY-AWARE PIPELINE FOR TOOTH DETECTION, FDI NUMBERING, SEGMENTATION AND FINDING CLASSIFICATION ON PANORAMIC DENTAL RADIOGRAPHS"

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are bonafide students of this institute and the work has been carried out by them under the supervision of **Dr. Pradnya Narkhede** and it is approved for the partial fulfillment of the requirement of Pimpri Chinchwad College of Engineering, an autonomous institute, for the award of the B. Tech. degree in Computer Engineering.

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Parts of this report, including the drafting of text and the drawing of the schematic diagrams, were prepared with the assistance of an AI system (Claude, Anthropic), working from the project's own source code, test records and evaluation reports. The team checked the content against those sources and takes responsibility for the report.

Finally, we thank our families and friends for their patience during the long training runs and evaluations.

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ABSTRACT

Reading a panoramic dental radiograph means locating every tooth, giving it its FDI number and checking it for findings. DentaMind AI is a research prototype, built on the public DENTEX dataset, that performs these steps with trained models and then uses a generative step to turn the structured model output into readable draft reports.

The system chains a Faster R-CNN tooth detector, a second Faster R-CNN for FDI numbering, a Mask R-CNN instance segmenter and a ResNet-50 classifier that names the closest of four finding classes (Caries, Deep Caries, Impacted, Periapical Lesion) for each detected tooth. Normalized entropy and flip disagreement estimate uncertainty, and thresholds fitted on a calibration split route uncertain regions to a human reviewer. The generative component drafts English, Hindi and Marathi reports from the structured findings only, using a deterministic template by default and an optional language-model path guarded by an evidence validator.

On an internal test split of 63 radiographs, the detector reached AP50 0.966, the segmenter mask AP50 0.974 (mean matched Dice 0.873), and 96.8 % of matched teeth received the correct FDI number. On ground-truth crops from 73 test radiographs, the classifier reached accuracy 0.785 and macro-F1 0.646, but Periapical Lesion recall was only 0.308; with detector crops in place of ground-truth crops, validation macro-F1 fell from 0.718 to 0.634. The entropy flag marked 54.2 % of test crops, which held 86.8 % of the errors. Four classifier-improvement attempts and an active-learning study were negative. All 61 report-output checks passed, and the test suite passed 590 tests.

Splits are grouped by pixel hash and test-split use is logged, but patient-level leakage cannot be excluded. The classifier has no healthy class, the language-model path has never been run live, and the system has not been clinically validated.

Keywords: panoramic radiograph, DENTEX, tooth detection, FDI numbering, instance segmentation, uncertainty estimation, human review, evidence-grounded report generation, generative AI, data leakage.

TABLE OF CONTENTS

Sr. No.	Title of Chapter	Page No.
01	**Introduction**	1
	1.1 Overview of Generative AI	1
	1.2 Project Background	2
	1.3 Motivation	2
	1.4 Problem Statement	3
	1.5 Objectives	3
	1.6 Scope of the Work	4
02	**Literature Survey**	5
	2.1 Review of Recent Literature	5
	2.2 Generative AI: Existing Approaches	6
	2.3 Review of Related Generative AI Systems	6
	2.4 Existing Solutions and State-of-the-Art Methods	7
	2.5 Gap Identification	8
	2.6 Common Findings from Literature	8
03	**Requirements Specification**	10
	3.1 Hardware Requirements	10
	3.2 Software Requirements	10
	3.3 Functional Requirements	11
	3.4 Non-Functional Requirements	12
	3.5 Generative AI Model Requirements	13
	3.6 Dataset and Knowledge-Base Requirements	14
04	**System Design**	17
	4.1 Proposed System Architecture	17
	4.2 System Block Diagram	18
	4.3 System Workflow	20
	4.4 Mathematical Model	24
05	**Project Implementation**	28
	5.1 Overview of Project Modules	28
	5.2 Data Collection and Preprocessing	29
	5.3 Generative AI Model Selection	31
	5.4 Model Configuration and Fine-Tuning	32
	5.5 Project Modules	33
	5.6 Tools and Technologies Used	35

Sr. No.	Title of Chapter	Page No.
	5.7 Algorithm Details	35
06	**Software Testing**	38
	6.1 Test Cases	38
	6.2 Generative AI Output Testing	40
	6.3 Performance Evaluation	41
07	**Results and Discussion**	43
	7.1 Project Outcomes	43
	7.2 Result Analysis	44
	7.3 Generated Output Analysis	46
	7.4 Performance Metrics	48
	7.5 Accuracy, Relevance and Quality Evaluation	50
	7.6 Validation of Results	52
	7.7 Comparison with Existing Approaches	55
	7.8 Screenshots of the Implemented System	56
08	**Conclusion and Future Work**	60
	8.1 Conclusion	60
	8.2 Limitations of the Proposed System	60
	8.3 Future Work	61
	8.4 Applications	62
	References	63
	Appendix A: Publication details (paper, reviewer comments, certificates)	67
	Appendix B: Plagiarism report of the project report	78

LIST OF ABBREVIATIONS

ABBREVIATION	ILLUSTRATION
AI	Artificial Intelligence
API	Application Programming Interface
AP	Average Precision
AP50	Average Precision at an IoU match threshold of 0.5
AUC	Area Under the (ROC) Curve
AUROC	Area Under the Receiver Operating Characteristic curve
CNN	Convolutional Neural Network
COCO	Common Objects in Context (dataset)
CSV	Comma-Separated Values
CUDA	Compute Unified Device Architecture
DENTEX	Dental Enumeration and Tooth Pathosis Detection Benchmark
ECE	Expected Calibration Error
FDI	Fédération Dentaire Internationale (two-digit tooth notation, ISO 3950)
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
GT	Ground Truth
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IoU	Intersection over Union
JS	Jensen-Shannon (divergence)
JSON	JavaScript Object Notation
LLM	Large Language Model
mAP	mean Average Precision
MDM	Multidisciplinary Minor
NLG	Natural Language Generation
NMS	Non-Maximum Suppression
PCCOE	Pimpri Chinchwad College of Engineering
PNG	Portable Network Graphics
R-CNN	Region-based Convolutional Neural Network
RAG	Retrieval-Augmented Generation

ABBREVIATION	ILLUSTRATION
ResNet	Residual Network
RGB	Red, Green, Blue (colour channels)
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SHA-256	Secure Hash Algorithm, 256-bit
SQL	Structured Query Language
TP	True Positive
TPR / FPR	True Positive Rate / False Positive Rate
TTA	Test-Time Augmentation
TXT	Plain-text file
UI	User Interface

NOMENCLATURE

SYMBOL	MEANING
A, B	Two boxes or binary masks being compared
IoU(A, B)	Intersection over union of A and B
Dice(A, B)	Dice coefficient of A and B
K	Number of finding classes (K = 4)
N	Number of training crops (N = 2,520)
n_c	Number of training crops of class c
w_c	Loss weight of class c
z_k	Logit (pre-softmax score) of class k
p_k	Softmax probability of class k
p, **q**	Softmax vectors of a crop and of its horizontally flipped copy
H(p)	Normalized predictive entropy
H′	Unnormalized (Shannon) entropy
JS(p, q)	Jensen-Shannon divergence between p and q, normalized by ln 2
u_{mask}	Mask instability of a region under horizontal flipping
S	Image-level combined uncertainty score
α_j, s_j	Weight and value of uncertainty signal j
τ_H, τ_{JS}, τ_M, τ_D	Review thresholds: entropy, flip disagreement, mask instability, detection score
J	Youden's index, J = TPR − FPR
T	Temperature used in temperature scaling
B_m	m-th confidence bin used by the ECE
acc(B_m), conf(B_m)	Accuracy and mean confidence of bin m
y_{ik}	One-hot ground-truth label of sample i for class k
BS	Multi-class Brier score
P, R, F1	Precision, recall and their harmonic mean
cov, risk	Coverage and error rate of an abstention rule

LIST OF FIGURES

FIGURE	ILLUSTRATION	PAGE NO.
4.1	System architecture	18
4.2	System block diagram	19
4.3	End-to-end processing workflow	20
4.4	Detection, FDI numbering and segmentation	21
4.5	Finding-classification workflow	22
4.6	Uncertainty and review routing	23
4.7	Evidence-grounded report generation	24
5.1	Dataset splitting and integrity protocol	30
7.1	Experimental evaluation framework	44
7.2	X-ray Analysis page before analysis	57
7.3	Analysis result with numbered regions	57
7.4	English research-draft report	58
7.5	Marathi view of a report	58
7.6	Region review with second approver	59
7.7	Uncertainty page	59

LIST OF TABLES

TABLE	ILLUSTRATION	PAGE NO.
2.1	Comparison of Related Studies with DentaMind AI	5
2.2	Generative Approaches and Their Suitability for Evidence-Grounded Dental Reports	6
2.3	Gaps in Existing Work and the Response in This Project	8
3.1	Hardware Used for Development, Training and Testing	10
3.2	Software Stack (Installed Versions)	11
3.3	Functional Requirements and Their Status	11
3.4	Non-Functional Requirements	13
3.5	Requirements for the Report-Generation Component	14
3.6	DENTEX Sub-Sets Present in the Project (Dataset Audit, 2026-10-14)	15
3.7	Split Statistics per Task	15
4.1	Review Thresholds Deployed in the Application	23
5.1	Main Modules and Their Location in the Code	28
5.2	Options Considered for the Generative Component	31
5.3	Training Configuration of the Four Deployed Checkpoints	32
5.4	Tools and Technologies Used	35
6.1	Acceptance Test Cases and Observed Outcomes (2026-10-14)	38
6.2	Summary of Software Test Results	39
6.3	Selected Report-Generation Test Cases (2026-10-14)	40
6.4	Inference Benchmark (RTX 3050 Laptop GPU, 10 Images)	41
7.1	Tooth Detection and Instance Segmentation	45
7.2	FDI Tooth Numbering	45
7.3	Finding Classification on Ground-Truth Crops	46
7.4	Confusion Matrices of the Finding Classifier (Rows: True Class; Validation / Test)	46
7.5	Excerpt of a Generated English Report (Exported 2026-10-14)	47
7.6	Summary of Test Metrics	48
7.7	Uncertainty and Calibration of the Finding Classifier	49
7.8	Flip-Instability Validation (30 Validation Radiographs)	49
7.9	Abstention Rules on Detector Crops	50
7.10	End-to-End Validation on Detector Crops (65 Radiographs, 331 Findings)	51

TABLE	ILLUSTRATION	PAGE NO.
7.11	Leakage Checks (Held-Out Test Split)	52
7.12	Recorded Uses of the Internal Test Split	53
7.13	Classifier-Improvement Experiments (Validation, 5 Seeds)	54
7.14	Active Learning: Validation Macro-F1 (Mean ± SD, 3 Seeds)	55
7.15	Comparison of Evaluation Settings with Existing Approaches	56
A.1	Status of Publication-Related Documents	67

CHAPTER 1
INTRODUCTION

1.1 OVERVIEW OF GENERATIVE AI

Generative artificial intelligence refers to models that learn the structure of a body of data well enough to produce new data of the same kind: images, text, audio or structured records. Three families dominate current practice. Generative adversarial networks train a generator against a discriminator until the generated samples become hard to tell apart from real ones [1]. Diffusion models learn to reverse a gradual noising process and produce samples by denoising step by step [2]. Transformer language models, built on self-attention [3], generate text one token at a time, and at large scale they can follow instructions and solve new tasks from a few examples [4].

Large language models have already been tested on medical questions. Singhal *et al.* showed that such models encode a substantial amount of clinical knowledge [5], and Thirunavukarasu *et al.* reviewed both the opportunities and the risks of using them in medicine [6]. The main risk for this project is hallucination: fluent text that is not supported by the input [7]. Retrieval-augmented generation reduces the problem by conditioning the generator on retrieved documents [8], but it does not remove it. In a clinical setting, a sentence that names the wrong tooth or invents a finding is worse than no sentence at all.

DentaMind AI uses generative AI in one clearly bounded place. The perception models that locate, number, segment and classify teeth are discriminative networks; they are not generative and the report does not describe them as such. The generative component is the reporting layer, which turns the structured output of those models into readable draft reports in English, Hindi and Marathi. It is built so that it can only restate what the models produced: every draft is checked against the structured evidence, and text that adds a tooth, a number, a finding name or a diagnostic phrase is rejected. Section 3.5 lists the requirements this places on the generator, and Section 5.3 explains why this conservative design was chosen.

1.2 PROJECT BACKGROUND

A panoramic radiograph shows both dental arches, the supporting bone and the temporomandibular region in a single exposure, which makes it one of the most common images in a dental examination. Reading it takes time. The reader has to find every tooth, assign it a two-digit FDI number (ISO 3950 [9]) and look at each tooth for findings such as caries, impaction or a periapical lesion. Convolutional neural networks have been studied for many of these tasks [10], from early tooth detection and numbering [11] to recent multi-stage systems that combine segmentation and numbering [12], [13].

The public DENTEX dataset made it possible to study enumeration and diagnosis on the same panoramic images. Its annotations are hierarchical: quadrants, then teeth with their quadrant and index, then teeth that carry one of four findings [14]. The DENTEX authors also published a diffusion-based hierarchical detector for the combined task [15], which is itself an example of a generative model used for perception.

The DentaMind code base started from the open-source CCB computer vision repository, which provides template notebooks for classification, detection, instance segmentation and Grad-CAM on the DENTEX data [16]. Those research notebooks were kept unchanged. The team added everything that makes up the system described in this report: the core library, the FastAPI back end, the React front end, the configuration, the training and evaluation scripts, the report generator and the test suite.

1.3 MOTIVATION

Three observations motivated the project. The first is that published results for dental AI are often reported per task, while a usable tool has to run the tasks in sequence. A finding classifier is usually scored on hand-drawn tooth boxes, but in a real system it receives whatever the detector returns, so detection errors propagate. We wanted to measure that effect rather than assume it away.

The second is that a model which is wrong without warning is hard to use safely. A tool that marks the regions it is unsure about, and hands them to a person, can still be useful even when its accuracy is modest, provided the uncertainty signals actually separate correct from incorrect outputs. This had to be tested, not claimed.

The third concerns the written output. Clinicians and patients read reports, not tables of scores, and in Maharashtra a report in Hindi or Marathi can be easier to follow than one in English. Generative AI makes fluent drafting cheap, but it also makes confident errors cheap. The project therefore needed a way to produce readable multilingual drafts that cannot say more than the models found.

1.4 PROBLEM STATEMENT

To design, implement and evaluate a research system that takes a panoramic dental radiograph and (i) detects, numbers and segments the teeth, (ii) names the closest of the four DENTEX finding classes for each detected tooth, (iii) estimates the uncertainty of each region and routes uncertain regions to a human reviewer, and (iv) generates evidence-grounded draft reports in English, Hindi and Marathi that cannot introduce content absent from the model output. The system has to be evaluated on held-out data with a documented protocol that prevents duplicate radiographs from leaking across splits, and its limitations have to be reported as clearly as its results.

1.5 OBJECTIVES

1. To audit the DENTEX data, remove the effect of duplicated radiographs, and build train, validation, calibration and test splits grouped by pixel content.
2. To train and evaluate a tooth detector, an FDI numbering model, a tooth instance segmenter and a four-class finding classifier, and to link their outputs region by region through IoU matching.
3. To measure how finding classification changes when the classifier receives detector crops instead of ground-truth crops.
4. To compute per-region uncertainty signals, fix review thresholds on a separate calibration split, and test whether the signals capture the classifier’s errors.
5. To build an evidence-grounded report generator with English, Hindi and Marathi output, an automatic validator and named-reviewer sign-off.
6. To deliver a working local application (FastAPI back end and React front end) with human review, and to verify it with automated and acceptance tests.

7. To report negative results, unfavourable metrics and data-integrity limitations alongside the positive findings.

1.6 SCOPE OF THE WORK

The work covers adult panoramic radiographs from the DENTEX dataset, the four finding classes that DENTEX annotates, and a locally deployed prototype that runs on a single workstation. Within that scope it includes data auditing, model training and evaluation, uncertainty estimation and review routing, evidence-grounded report drafting in three languages, a human-review workflow with a second approver, an active-learning study and a full software test campaign.

Several things are deliberately outside the scope. The system is not a diagnostic device and makes no clinical claim; no clinical study has been carried out. The classifier has no healthy class, so it cannot declare a tooth healthy. Pediatric and mixed-dentition images, other imaging modalities and other datasets were not studied. The optional language-model path for rewording reports is implemented but was never connected to a live model, and the Hindi and Marathi wording has not been reviewed by a native-speaking dental professional. These limits are restated in Chapter 8.

CHAPTER 2
LITERATURE SURVEY

2.1 REVIEW OF RECENT LITERATURE

Deep learning for dental radiographs has moved from single tasks to multi-stage pipelines. Schwendicke *et al.* mapped the early use of convolutional networks for dental image diagnostics in a scoping review [10]. Tuzoff *et al.* detected and numbered teeth on panoramic radiographs with a Faster R-CNN detector followed by a classification network [11]. More recently, Putra *et al.* detected and numbered permanent teeth with a deep detector [12], and Zhang *et al.* built a multi-stage system that couples instance segmentation with FDI-compliant numbering and treats missing and supernumerary teeth explicitly [13].

For findings, Szabó *et al.* evaluated a commercial system for periapical lesions on 616 teeth from 357 panoramic radiographs [17], and Bittar *et al.* summarized the evidence on AI for caries detection in an umbrella review [18]. Hamamci *et al.* released DENTEX [14] together with a diffusion-based hierarchical detector for its enumeration-and-diagnosis task [15]. Two lessons from this literature shaped the project: results depend strongly on how the data were split, and rare findings such as periapical lesions are much harder than common ones. Table 2.1 places these studies next to DentaMind AI.

Table 2.1: Comparison of Related Studies with DentaMind AI

Study	Task and data	Relation to this project
Schwendicke *et al.* [10]	Scoping review of CNNs for dental image diagnostics	Background on early CNN work in dental imaging
Tuzoff *et al.* [11]	Tooth detection and numbering, panoramic radiographs	Same tasks; we add segmentation, findings and uncertainty
Putra *et al.* [12]	Permanent-tooth detection and numbering, panoramic	Same tasks; we add end-to-end evaluation
Zhang *et al.* [13]	Multi-stage instance segmentation and FDI numbering	Same task pair; we link stages by IoU and add review routing
Szabó *et al.* [17]	Commercial periapical-lesion detection, 357 panoramic radiographs	Periapical Lesion is our weakest class
Bittar *et al.* [18]	Umbrella review of AI caries detection	Context for the Caries and Deep Caries results
Hamamci *et al.* [14], [15]	DENTEX benchmark; diffusion-based hierarchical detector	Source of our data and finding taxonomy

2.2 GENERATIVE AI: EXISTING APPROACHES

Generative models differ in what they generate and in how much control the user has over the output. Adversarial networks [1] and diffusion models [2] generate images and have been used in medical imaging for synthesis and, in the DENTEX baseline, for detection [15]. Transformer language models [3], [4] generate text and can describe structured data in natural language. Retrieval-augmented generation adds documents retrieved at run time to the model's input, so that the text can be grounded in a knowledge source [8]. Older data-to-text systems generate text from structured records with explicit document plans and sentence templates; they are less fluent but fully predictable [19].

Table 2.2 compares these approaches for the specific job DentaMind needs done: turning a list of model findings into a short, faithful report. The deciding factor was not fluency but control. A report generator in this setting must never add a tooth, a number or a diagnosis that the models did not produce.

Table 2.2: Generative Approaches and Their Suitability for Evidence-Grounded Dental Reports

Approach	What it generates	Main risk for reports	Use in DentaMind AI
GAN [1]	Images	Not designed for text; synthetic images could be mistaken for data	Not used
Diffusion model [2]	Images; also used for detection [15]	Heavy computation for a 4 GB GPU	Not used
Large language model [3], [4]	Fluent free text	Hallucinated teeth, numbers or diagnoses [7]	Optional rewording only, behind the validator; never run live
Retrieval-augmented generation [8]	Text grounded in retrieved documents	Retrieved text can still be misused; needs a curated corpus	Not used (no curated dental corpus)
Template-based data-to-text [19]	Text from structured records	Less fluent; wording must be written by hand	Default generator in English, Hindi and Marathi

2.3 REVIEW OF RELATED GENERATIVE AI SYSTEMS

Automatic report generation for medical images has its own line of work. Jing *et al.* generated medical imaging reports, including chest X-ray reports, with a co-attention model that first predicts tags and then writes the text [20]. Chen *et al.* introduced R2Gen, a memory-driven Transformer that writes complete radiology reports directly from images [21]. These systems learn from large collections of paired images and reports.

DENTEX contains no reports at all, so this approach was not available to us even in principle.

General-purpose biomedical models point in the same direction. Singhal *et al.* reported strong results for the Med-PaLM models on medical question-answering benchmarks, while their human evaluation still found gaps relative to clinicians [5], and Tu *et al.* extended the idea to a single model that handles images, text and genomics [22]. Both papers are careful to note failure modes, and the survey by Ji *et al.* shows that hallucination is a property of neural text generation in general rather than of any one model [7]. For a student project without clinical reviewers, these results argue for keeping the generator on a short leash: let it rephrase, never let it decide.

2.4 EXISTING SOLUTIONS AND STATE-OF-THE-ART METHODS

2.4.1 Detection, Segmentation and Classification

Faster R-CNN proposes candidate regions with a region-proposal network and classifies them in a second stage [23]. A feature pyramid network gives the detector features at several scales [24], which suits panoramic images, where incisors and molars differ greatly in size. Mask R-CNN adds a mask branch to Faster R-CNN and produces one binary mask per instance [25]. Residual networks make very deep classifiers trainable through identity shortcuts [26]. In practice all of these are initialized from weights pre-trained on large general-purpose datasets, COCO for detection [27] and ImageNet for classification [28], and then fine-tuned on the target data.

2.4.2 Uncertainty, Calibration and Selective Prediction

Huang *et al.* review uncertainty quantification for medical image analysis and distinguish probabilistic from non-probabilistic methods [29]. Test-time augmentation gives an inexpensive estimate from a single network by comparing predictions on transformed copies of the input [30]. Selective classification lets a model abstain when it is unsure, trading coverage for lower risk [31]. Modern networks tend to be over-confident; temperature scaling is the standard post-hoc correction [32], and the expected calibration error [33] and the Brier score [34] are the usual measures. Youden's index [35] is a common way to pick a decision threshold, and the Jensen-Shannon divergence [36] measures how far two probability vectors differ.

2.4.3 Explainability, Active Learning and Data Integrity

Grad-CAM produces a coarse heat map of the image regions that drive a classifier's decision [37]. Active learning tries to reduce labeling effort by choosing which images to annotate next, and Wang *et al.* survey its use in medical imaging [38]. On the data side, Kapoor and Narayanan show that leakage between training and test data is a common cause of irreproducible machine-learning results [39], and Huang *et al.* measured how near-duplicate images inflate a brain-MRI benchmark [40]. These two papers explain why so much of this project's effort went into the data protocol of Section 5.2.

2.5 GAP IDENTIFICATION

Reading the literature against the needs of a usable tool shows five gaps. Table 2.3 lists them with the way the project responds. The responses are evaluations and safeguards rather than new network architectures; the contribution of this project is evidence about how a complete pipeline behaves, not a new model.

Table 2.3: Gaps in Existing Work and the Response in This Project

Gap	Response in DentaMind AI
Stages are evaluated separately; classifiers are scored on ground-truth boxes	An end-to-end study on detector crops (Section 7.5)
Uncertainty is rarely tied to an operational review decision	Thresholds fixed on a calibration split, then tested once on held-out data (Section 7.4)
Generated reports can hallucinate content	Template generation by default, an evidence validator for every draft, LLM output rejected on any violation (Section 6.2)
Duplicate radiographs and repeated test use inflate results	Pixel-hash grouped splits, leakage checks and a hash-chained test-set ledger (Section 7.6)
Negative results are seldom reported	Four improvement attempts and an active-learning study reported with their failed criteria (Section 7.6)

2.6 COMMON FINDINGS FROM LITERATURE

- Detecting and numbering teeth on panoramic radiographs is now reliable on curated data, while recognizing findings, especially rare ones, remains much harder.
- Most models are fine-tuned from general-purpose pre-trained weights, and results depend heavily on dataset composition and splitting.

- Uncertainty estimates are useful only if they are checked against actual errors on data that were not used to tune them.
- Neural text generators are fluent but can produce unsupported statements, so medical text generation needs grounding and checking.
- Leakage and duplicate images are a recognized source of inflated performance, and patient-level grouping requires identifiers that public datasets often lack.

CHAPTER 3
REQUIREMENTS SPECIFICATION

This chapter lists what the system needs and what it must do. Every requirement marked as implemented was checked against the code and the project's test records; items that are optional or were not exercised are marked as such, so that the reader can separate the delivered system from possible extensions.

3.1 HARDWARE REQUIREMENTS

All training, evaluation and testing were done on one laptop, whose configuration is given in Table 3.1. The values were read from the machine itself and agree with the hardware named in the training records. A CUDA-capable GPU is needed for practical training. Inference also runs on the CPU, at a cost measured during the acceptance tests.

Table 3.1: Hardware Used for Development, Training and Testing

Component	Configuration used	Notes
Processor	Intel Core i5-12450H (12th generation), 8 cores, 12 threads	CPU-only inference took about 7 to 23 s per prediction and 11 to 53 s per full analysis
GPU	NVIDIA GeForce RTX 3050 Laptop GPU, 4 GB	Training both detectors at full resolution needs about 3 GB of GPU memory; one training job at a time
Memory	15.7 GB RAM	Configuration used; no minimum was measured
Storage	DENTEX training data: 3,606 files, 14.7 GB; unlabeled validation data: 51 files, 0.2 GB	Additional space for checkpoints, the SQLite database and reports
Operating system	Windows 11 Home (64-bit)	Windows scripts provided; the Docker files were not tested

Sources: the development machine (queried 2026-10-26), docs/FINAL_ACCEPTANCE_REPORT.md (CPU timings, data sizes) and docs/KNOWN_LIMITATIONS.md (GPU memory).

3.2 SOFTWARE REQUIREMENTS

Table 3.2 gives the software stack with the versions installed in the project environment. The back end and the models are written in Python; the user interface is a single-page React application built with Vite. Components that are supported by the code but were not installed or configured are listed separately, because the report must not imply that they were used.

Table 3.2: Software Stack (Installed Versions)

Layer	Software	Version
Language	Python	3.13.0
Deep learning	PyTorch (CUDA 12.4 build); torchvision	2.6.0; 0.21.0
Numerics and images	NumPy; Pillow	2.5.4; 12.3.0
Web API	FastAPI; Uvicorn; Pydantic	0.143.0; 0.54.0; 2.14.1
Database	SQLite (Python sqlite3 module)	3.45.3
Front end	React; React Router; Recharts; lucide-react	18.3.1; 6.30.6; 2.15.4; 0.460.0
Build and styling	Vite; Tailwind CSS	5.4.21; 3.4.19
Testing	pytest (also runs the unittest suites)	9.1.1
Optional, not configured	OpenAI-compatible LLM endpoint; Label Studio; TensorBoard; Docker	not installed or not run

Versions read from the project's .venv and frontend/node_modules on 2026-10-26.

3.3 FUNCTIONAL REQUIREMENTS

The functional requirements follow the path of one radiograph through the system, from upload to a signed-off report, and add the research functions used to evaluate the models. Table 3.3 gives each requirement with the evidence that it is met; acceptance test identifiers (AT) refer to Table 6.1 in Chapter 6.

Table 3.3: Functional Requirements and Their Status

ID	Requirement	Status and evidence
FR-01	Accept PNG or JPEG radiographs up to 25 MB; check extension and file signature; reject anything else without storing it	Implemented; AT-04, AT-05
FR-02	Detect every tooth and return its box and score	Implemented; AT-06
FR-03	Assign an FDI number to each detected tooth, or leave it unassigned; never invent a number when the model is missing	Implemented; AT-10
FR-04	Segment tooth instances and link each mask to one detected tooth	Implemented; AT-08
FR-05	Name the closest of the four finding classes for each tooth and show the raw score	Implemented; AT-07
FR-06	Compute uncertainty signals per region; report a missing signal as unavailable, never as zero	Implemented; AT-11
FR-07	Flag uncertain or problematic regions for review and order images by a combined score	Implemented; AT-11, review tests
FR-08	Show a Grad-CAM heat map for a selected region	Implemented; verified through the API, not clicked in the browser
FR-09	Let a reviewer confirm, correct or reject a region, keep a revision history, and require a second named approver	Implemented; review workflow tests (27 of 27)

ID	Requirement	Status and evidence
FR-10	Generate a draft report from the structured findings and validate it	Implemented; AT-12
FR-11	Show the same report in Hindi and Marathi without changing any structured value	Implemented; AT-13
FR-12	Export reports as HTML, TXT or JSON and reopen saved reports	Implemented; AT-14, AT-15
FR-13	Register checkpoints with SHA-256 checks and link them to their evaluation reports	Implemented; AT-16
FR-14	Score the unlabeled pool and select images for annotation by random, entropy or combined strategies, never from the test split	Implemented; unit and API tests
FR-15	Start training and validation-split evaluation jobs from the API	Partly verified: smoke runs only

3.4 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements matter as much as the functional ones, because the main risk of a medical prototype is not a crash but a confident wrong answer. Table 3.4 lists them.

Table 3.4: Non-Functional Requirements

Quality	Requirement	How it is met
Honesty of output	No invented predictions, numbers or signals	Missing model gives HTTP 503; unavailable signals are labelled; validator rejects unsupported report text
Integrity	Models and test data cannot be swapped or reused silently	SHA-256 check of every checkpoint against its sidecar; test split locked and every use logged
Privacy	Radiographs stay on the machine	Local processing; stored names derived from content hashes; the optional LLM would receive only de-identified findings
Security	Only the local user can call the API	API token, loopback-only access, per-client rate limits (23 live checks passed)
Performance	Interactive analysis on a laptop GPU	Full analysis of one image in 1.92 s in the UI audit; server ready 4.1 s after start
Reliability	A failing step must not hide the others	Each of the ten steps reports ok, unavailable or failed; no HTTP 500 in the audited access log
Usability	Clear, multilingual output with visible limitations	English, Hindi and Marathi views; disabled controls state the reason
Maintainability	Changes are covered by automated tests	590 tests passed on 2026-10-26
Reproducibility	Results can be traced to data and code	Seed 42, dataset fingerprint, split hashes and software versions stored in every evaluation report
Portability	Runs without a GPU	Automatic CPU fallback, at the cost shown in Table 3.1

3.5 GENERATIVE AI MODEL REQUIREMENTS

The report generator is the only generative component, and its requirements are written to limit what it can say. They follow from the hallucination risk discussed in Chapter 2 [7] and from the fact that no clinician was available to review the generated text. Table 3.5 lists them together with the status of each.

Table 3.5: Requirements for the Report-Generation Component

Requirement	Implementation	Status
Use only structured model output as input	Evidence bundle with finding id, FDI code, class label, raw score, uncertainty and review status per region	Implemented and tested
Deterministic default	Template generator; regenerating a report gives identical findings	Implemented; RPT-A03
Never add unsupported content	Validator checks fields, FDI validity, scores, and scans the text for teeth, numbers, finding names and clinical phrases not in the evidence	Implemented; RPT-E01 to E07
No diagnostic or treatment language	Forbidden phrases include diagnosis, prescription, recommendation, treatment plan, extraction, root canal, antibiotic, “confirmed” and “patient has”	Implemented; RPT-C07, C08
Optional LLM rewording	OpenAI-compatible chat-completions call, temperature 0, JSON output, restrictive system prompt	Implemented; stub-tested only; never run live
De-identification	Only finding fields and a hashed image alias leave the machine; no image, path or original id	Implemented; unit test
Safe fallback	Invalid or non-JSON LLM output is discarded and the template draft is used, with the reasons recorded	Implemented; unit test
Three languages, identical values	Hindi and Marathi from fixed sentence templates; export refused if any value differs from English	Implemented; RPT-C01 to C06
Human sign-off	Every report is an unconfirmed research draft until a named reviewer signs it	Implemented; RPT-F01 to F04
Not required, not implemented	Fine-tuning of a language model; retrieval from a knowledge base; editing of drafts; PDF export	Out of scope

RPT identifiers refer to the report test campaign in docs/MULTILINGUAL_REPORT_TEST_REPORT.md (Section 6.2).

3.6 DATASET AND KNOWLEDGE-BASE REQUIREMENTS

3.6.1 Image Dataset

The models need panoramic radiographs with tooth boxes, polygon masks, FDI labels and finding labels. DENTEX provides all four [14]. Its radiographs come from three institutions and several scanners. Table 3.6 lists the sub-sets present in the project as measured by the dataset audit. The sub-sets overlap heavily: the 3,653 files contain only 2,716 distinct radiographs, and the 2,005 annotated entries only 1,133. As many as 235 radiographs appear in all three annotated sub-sets, sometimes under different file names, so DENTEX image counts are counts of entries, not of radiographs and certainly not of patients.

Table 3.6: DENTEX Sub-Sets Present in the Project (Dataset Audit, 2026-10-14)

Sub-set	Images	Annotations	Use in this project
quadrant	693	2,772	not used by the current models
quadrant-enumeration	634	18,095	detection, numbering, segmentation
quadrant-enumeration-disease	705	3,529	finding classification
unlabelled	1,571	n/a	active-learning pool
validation_data (unlabeled)	50	n/a	inference demonstration only
All files on disk	3,653		2,716 distinct radiographs by pixels

Source: Data/reports/dataset_audit.md. In the disease sub-set, 27 images have no annotation.

The disease sub-set contains 2,189 Caries, 578 Deep Caries, 604 Impacted and 158 Periapical Lesion annotations. There is no healthy label: a tooth without a finding is simply not annotated. Two parts of DENTEX are excluded from evaluation. The official test release uses a different, Turkish and treatment-oriented label set, no verified mapping to the four English classes was found, and 95 of its 250 images are byte-identical to radiographs in the training sub-sets. The 50 unlabeled *validation_data* images are used only to demonstrate inference; 20 of them duplicate labeled radiographs.

3.6.2 Split Requirements

Each split has a single purpose. Train fits the weights, validation selects the training epoch and gives validation metrics, calibration fixes the review thresholds and the temperature, and test is reserved for final metrics. Table 3.7 gives the resulting counts per task; the procedure is described in Section 5.2.

Table 3.7: Split Statistics per Task

Task (sub-set)	Train	Val.	Cal.	Test
Detection, numbering, segmentation: radiographs	440	69	62	63
Detection: ground-truth tooth boxes	n/a	2,000	n/a	1,790
Findings: radiographs	476	65	64	73
Findings: Caries crops	1,550	205	206	228
Findings: Deep Caries crops	438	45	47	48
Findings: Impacted crops	412	72	55	65
Findings: Periapical Lesion crops	120	9	16	13

Sources: dataset_audit.md, eval_*.json and reports/training_summary.md. The calibration split holds 324 finding crops; in training, Caries outnumber Periapical Lesion 12.9 to 1.

3.6.3 Knowledge Base for Report Generation

The report generator does not use a retrieval corpus or any external knowledge base. Its “knowledge” is a small controlled vocabulary kept in the code: the four class

names the classifier can output, the 32 valid FDI codes [9], the sentence templates for English, Hindi and Marathi, the standard disclaimer and limitation statements, and the list of forbidden clinical phrases used by the validator. Keeping this vocabulary closed is what allows the validator to reject any statement the models did not support.

CHAPTER 4
SYSTEM DESIGN

4.1 PROPOSED SYSTEM ARCHITECTURE

Figure 4.1 shows the architecture as implemented. A radiograph enters through the web interface and is passed unchanged to the back end. The tooth detector is the central model: its boxes are the regions that every later stage refers to. The numbering model, the segmenter and the finding classifier each attach information to those regions, and the uncertainty stage adds signals that decide whether a region needs review. Region records and the final result are stored in SQLite, from where the report service and the human-review service read them.

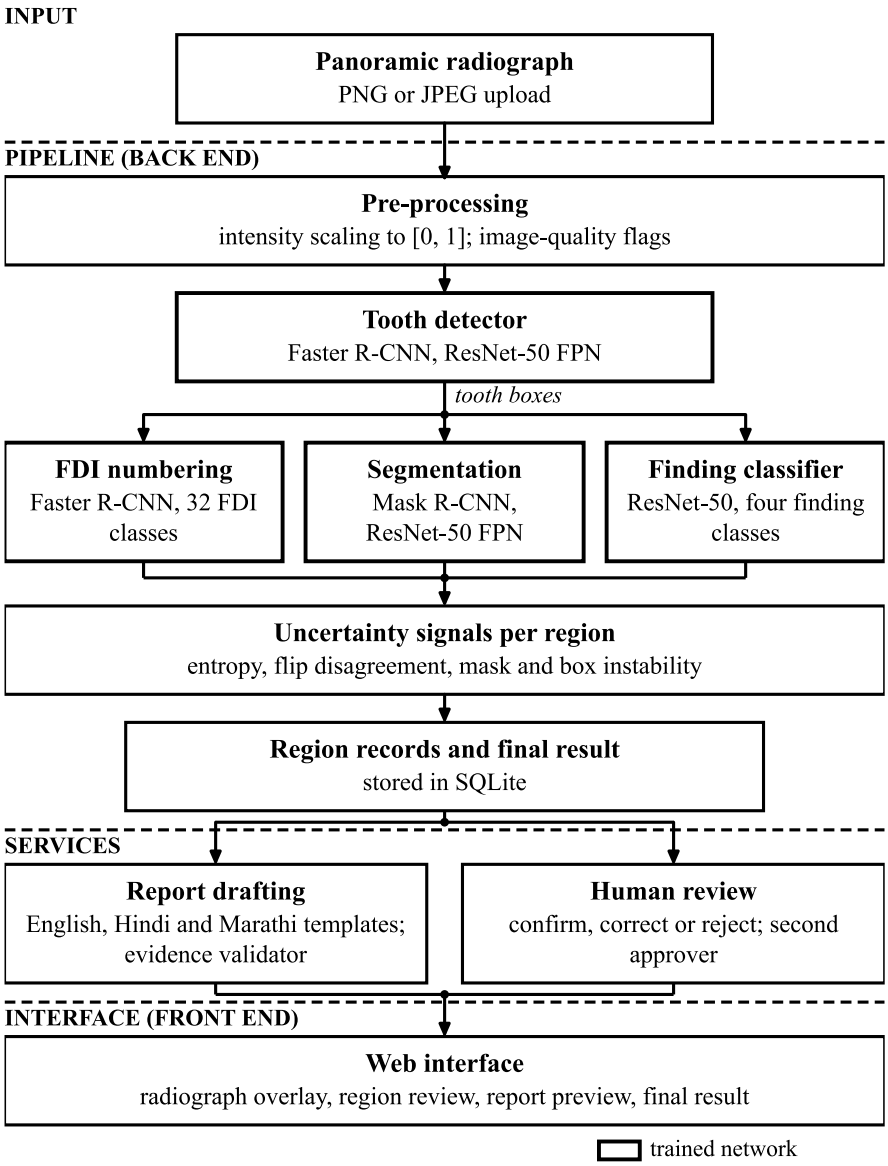


Figure 4.1: System architecture of DentaMind AI as implemented. Bold outlines mark trained networks.

Two design decisions shape this architecture. First, regions are linked between models by overlap (intersection over union), never by their position in a list, because the models return their outputs in different orders and some regions have no counterpart in another model. Second, the original upload is never modified; preprocessing works on a copy, so that a reviewer can always see the image the models analyzed.

4.2 SYSTEM BLOCK DIAGRAM

Figure 4.2 shows the same system as deployed software blocks. The browser talks only to the FastAPI back end, through the Vite development proxy that adds the API token, so the token never appears in the front-end code. The back end is split into services

for images, analysis, reports, reviews, active learning, models and the dataset. The model layer loads the four checkpoints through a registry that verifies each file's SHA-256 hash against its sidecar before use. Everything is stored on the local disk.

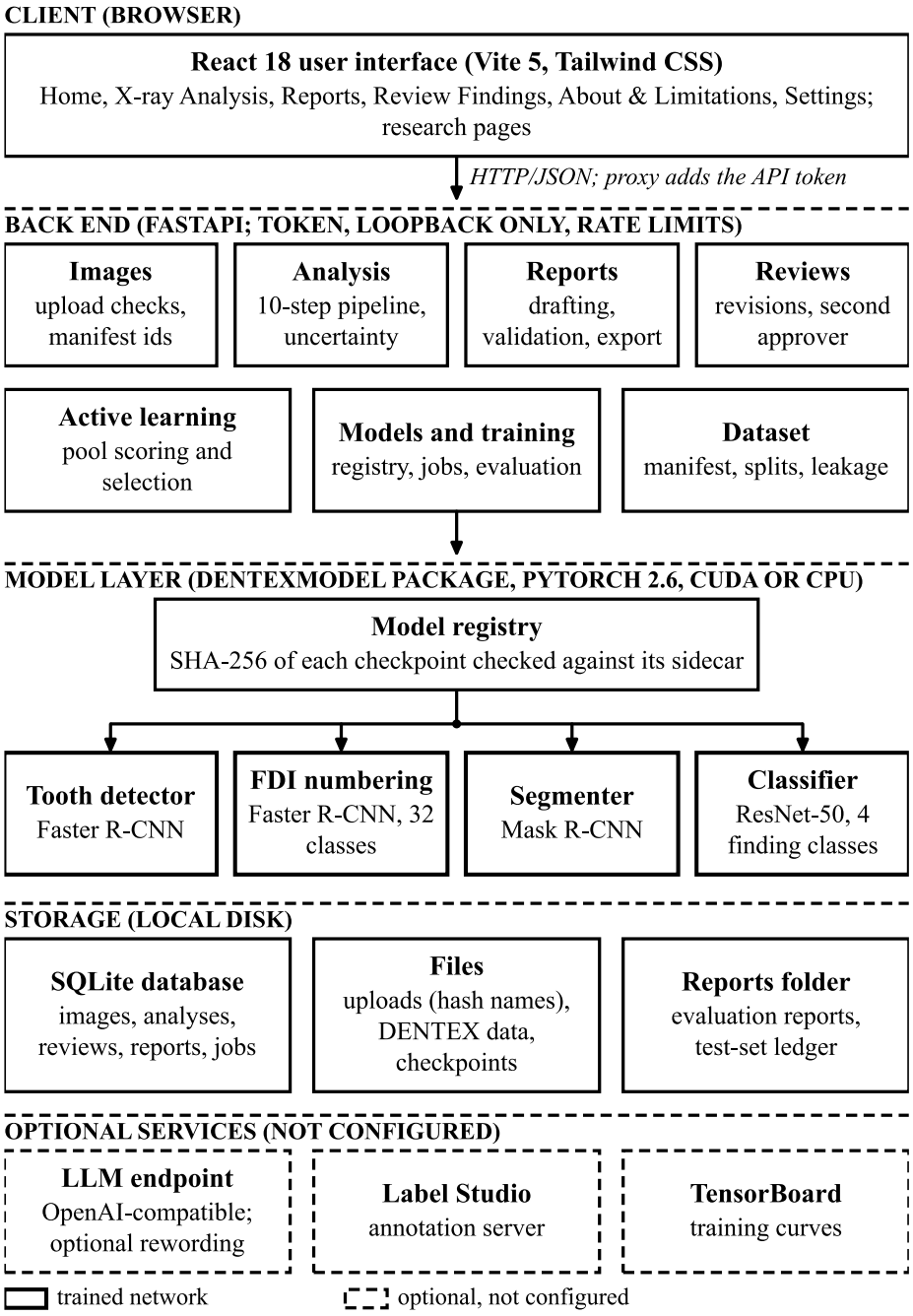


Figure 4.2: System block diagram: client, back-end services, model layer, storage and optional services (dashed, not configured).

The dashed blocks are optional services that the code supports but that were not configured in this deployment: an OpenAI-compatible language-model endpoint for

rewording reports, a Label Studio annotation server and TensorBoard. With the shipped configuration nothing leaves the machine.

4.3 SYSTEM WORKFLOW

4.3.1 End-to-End Processing

Figure 4.3 lists the ten pipeline steps in the order in which the analysis service runs them. Each step is timed and records its status. If a model is missing, its step is reported as unavailable; if a step crashes, it is reported as failed. Later steps that do not depend on it still run, and a missing signal is reported as unavailable instead of being set to zero, so that a reviewer can tell “no problem found” from “not measured”.

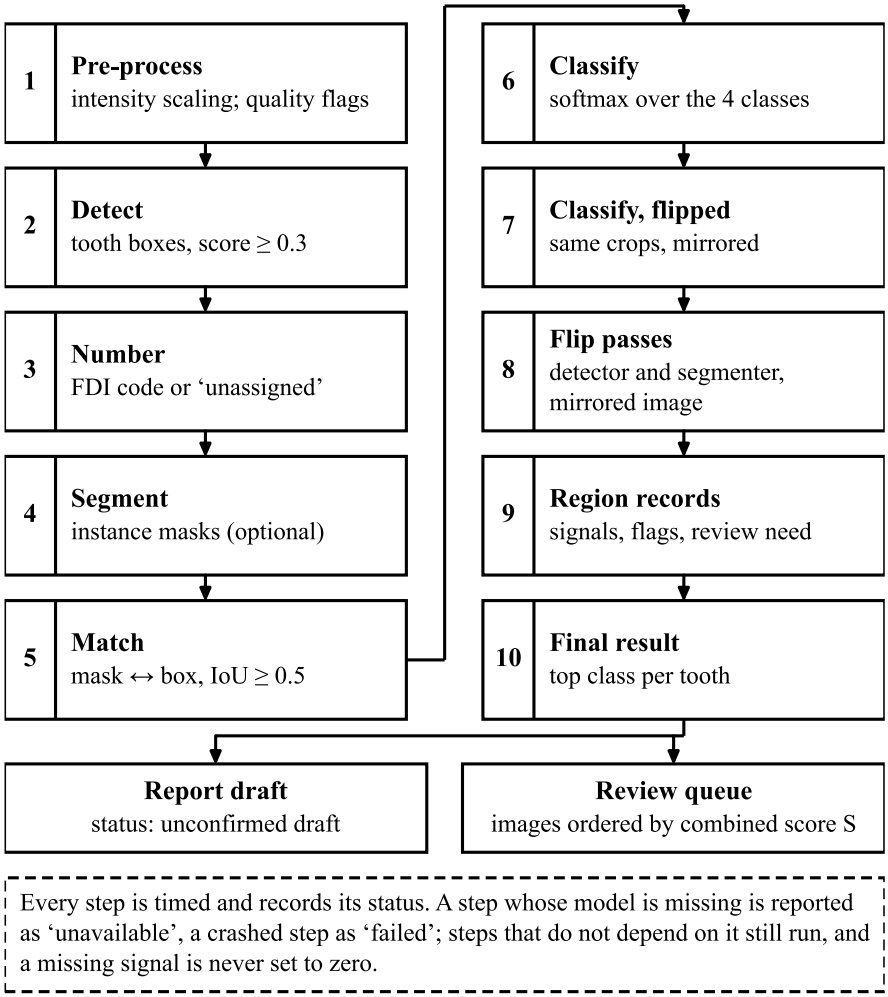


Figure 4.3: End-to-end processing workflow: the ten implemented pipeline steps, the two outputs, and (dashed) the behaviour when a step is unavailable or fails.

4.3.2 Detection, Numbering and Segmentation

Figure 4.4 shows how the three localization models are combined. The detector proposes tooth boxes with score at least 0.3. The numbering model is a second detector with one class per FDI code. Each detector box takes the unused numbering box with the highest overlap, provided the IoU is at least 0.5, and otherwise stays unassigned. Instance masks from the segmenter are matched one to one to detector boxes at the same threshold. A consistency check flags duplicated codes, low-confidence codes and codes on the wrong side or arch for their quadrant.

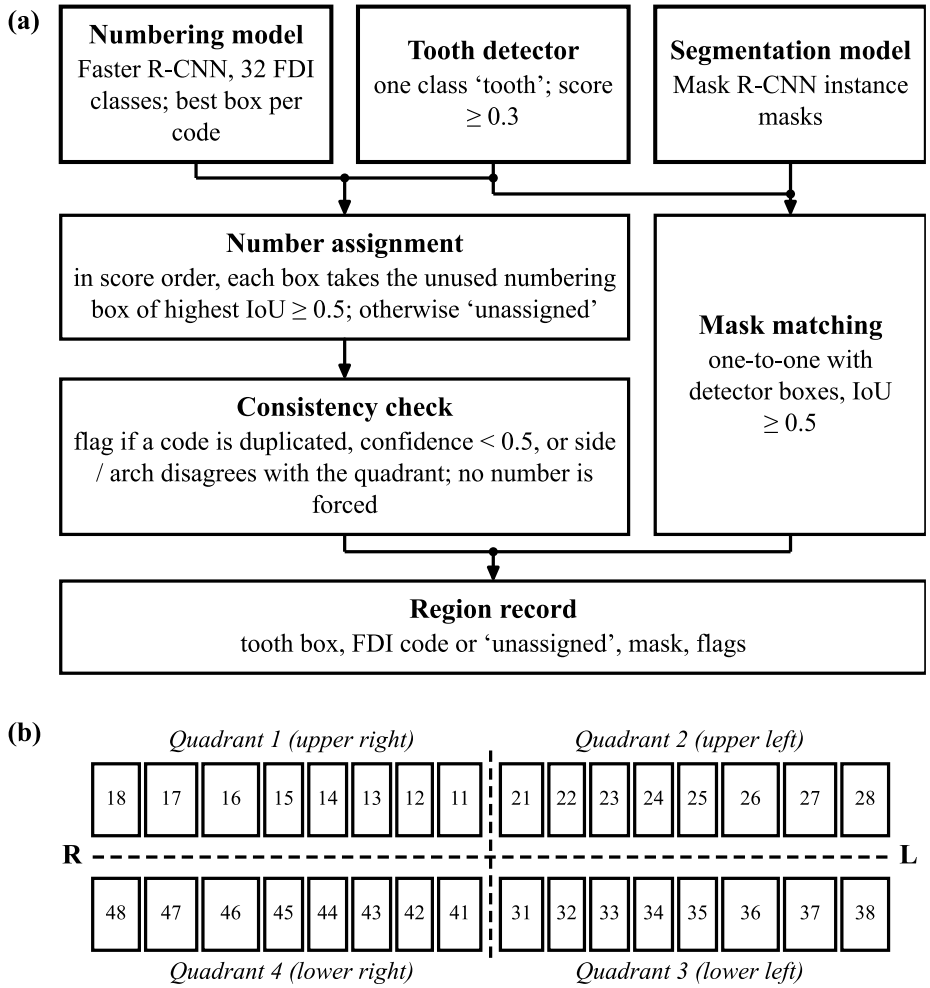


Figure 4.4: (a) Combination of detector boxes, numbering model and segmentation model into a region record. (b) FDI two-digit notation used for the 32 numbering classes, as the teeth appear on a panoramic radiograph.

4.3.3 Finding Classification

Figure 4.5 shows the classification path. Each tooth box is cropped with 10 % padding, resized to 224 × 224 pixels, normalized with ImageNet statistics and passed to a ResNet-50 with a four-way output layer. The application crops detector boxes, while

the evaluation of Chapter 7 crops ground-truth boxes; the difference between the two is measured in Section 7.5. Because DENTEX labels only teeth that carry a finding, the softmax output answers “which of these four findings does this crop most resemble”, not “is this tooth diseased”.

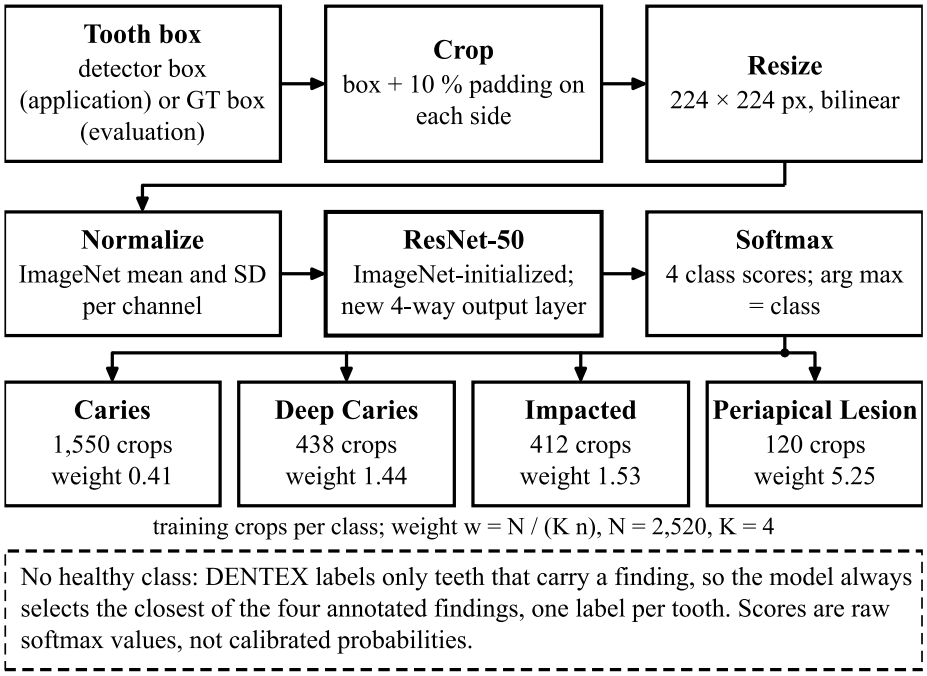


Figure 4.5: Finding-classification workflow with the training-crop count and class weight of each class; the dashed note states that there is no healthy class.

4.3.4 Uncertainty and Review Routing

Figure 4.6 separates what the application does from what was only evaluated offline. Five conditions send a region to review; their thresholds are listed in Table 4.1. The entropy and flip-disagreement thresholds were fixed once on the calibration split, using one seed and one dataset, and are therefore provisional. A region at or above the entropy threshold is marked as abstained. Since a project decision of 2026-10-16, its highest-scoring class is still shown, labelled as needing human review because of high classifier uncertainty, and an abstained tooth is never reported as healthy. Scores shown to users are raw softmax values, not calibrated probabilities.

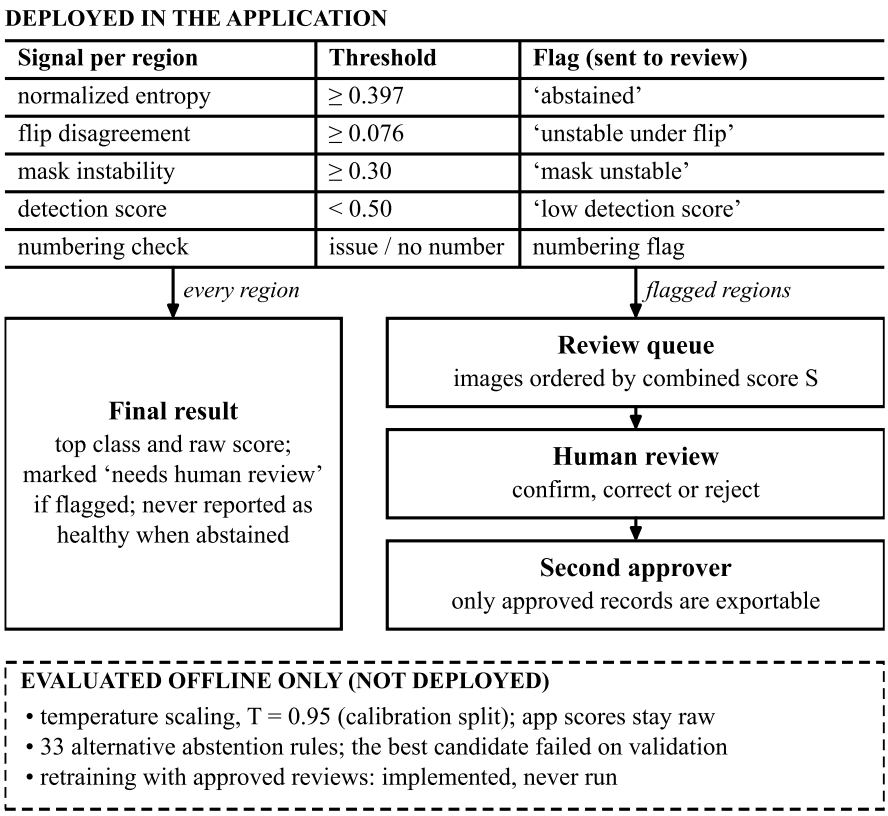


Figure 4.6: Uncertainty-aware review routing. Upper part: thresholds and routing deployed in the application. Lower dashed panel: methods evaluated offline only.

Table 4.1: Review Thresholds Deployed in the Application

Signal per region	Threshold	How it was set	Effect
Normalized entropy H	≥ 0.397	Maximizes Youden's index on calibration crops	abstained; review
Flip disagreement JS	≥ 0.076	90th percentile on calibration crops	review
Mask instability u_{mask}	≥ 0.30	Fixed design value	review
Detection score	< 0.50	Fixed design value	review
Numbering check	issue or no number	Rule-based consistency check	review

Sources: Data/reports/uncertainty_thresholds.json (entropy 0.3966 and flip disagreement 0.0762, chosen on the calibration split on 2026-10-14) and backend/services/pipeline.py (mask and detection thresholds; the built-in fallbacks 0.40 and 0.10 apply only if the threshold file is missing). Temperature scaling and alternative abstention rules were evaluated offline only.

4.3.5 Evidence-Grounded Report Generation

Figure 4.7 shows the generative component. A stored analysis is converted into an evidence bundle with one finding per region and a stable identifier (F1, F2, and so on). By default the template generator writes the summary and one sentence per finding. If a language-model endpoint were configured, the de-identified findings could be sent for rewording instead, but the returned text passes through the same validator, and any violation makes the system discard it and fall back to the template. The English draft is

the reference: the Hindi and Marathi views are produced from fixed sentence templates with all values copied verbatim, and an export is refused if any value differs. A report becomes a reviewed research draft only when a named person signs it.

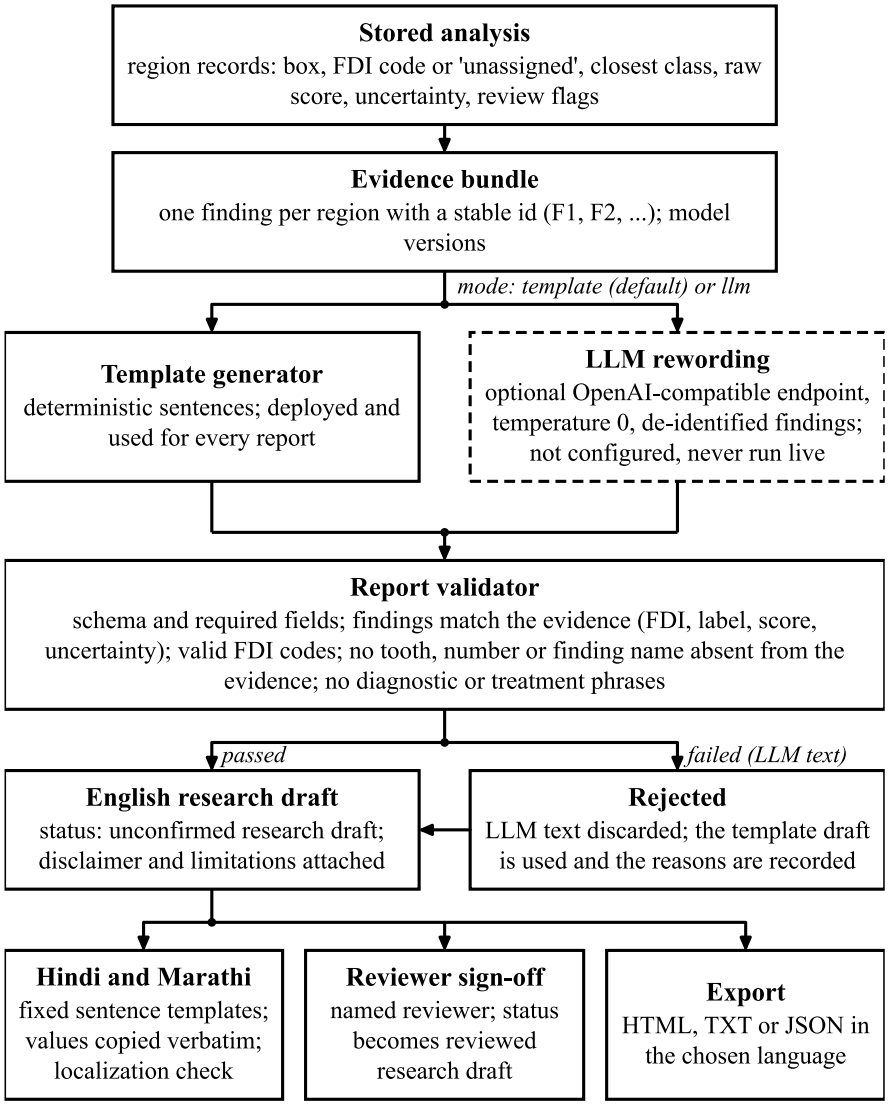


Figure 4.7: Evidence-grounded report generation as implemented; the dashed block (LLM rewording) is optional, not configured and never run against a live model.

4.4 MATHEMATICAL MODEL

This section gives the mathematical definitions used by the implemented system: the overlap measures that link regions, the class weights used in training, the uncertainty signals and the review rule. The evaluation measures follow in Subsection 4.4.5. All symbols are also listed in the Nomenclature.

4.4.1 Overlap Measures

Two regions A and B, which may be boxes or binary masks, are compared by their intersection over union and, for masks, by the Dice coefficient:

$$\mathrm{IoU}(A,B) = \frac{|A \cap B|}{|A \cup B|}, \quad \mathrm{Dice}(A,B) = \frac{2|A \cap B|}{|A| + |B|} \tag{4.1}$$

Two regions are matched when IoU ≥ 0.5. Matching is one to one and greedy: detector boxes are visited in descending score order and each takes the best unused partner.

4.4.2 Class Weights for the Finding Classifier

The training crops are strongly imbalanced, so the cross-entropy loss weights each class inversely to its frequency:

$$w_c = \frac{N}{K n_c} \tag{4.2}$$

where $N = 2{,}520$ training crops, $K = 4$ classes and n_c is the number of training crops of class c. With 1,550, 438, 412 and 120 crops this gives weights 0.41, 1.44, 1.53 and 5.25 for Caries, Deep Caries, Impacted and Periapical Lesion.

4.4.3 Uncertainty Signals

The classifier's logits z_k are turned into probabilities by the softmax. The normalized predictive entropy of a probability vector $\mathbf{p}$ over K classes lies between 0 and 1:

$$H(p) = -\frac{\sum_k p_k \ln p_k}{\ln K} \tag{4.3}$$

Flip disagreement compares the prediction $\mathbf{p}$ for a crop with the prediction $\mathbf{q}$ for its horizontally flipped copy, a two-pass form of test-time augmentation [30], using the Jensen-Shannon divergence [36] normalized by ln 2:

$$\mathrm{JS}(p,q) = \frac{H'\left(\frac{p+q}{2}\right) - \frac{H'(p)+H'(q)}{2}}{\ln 2} \tag{4.4}$$

where H' is the unnormalized Shannon entropy. Mask instability (and, in the same way, detection instability) measures how much an instance changes when the whole image is flipped, processed and flipped back:

$$u_{\mathrm{mask}} = 1 - \mathrm{IoU}\left(m, \tilde{m}_{\mathrm{flip}}\right) \tag{4.5}$$

where m is the mask and $\tilde{m}_{\text{flip}}$ its best match on the flipped image; an instance without a counterpart counts as 1. The image-level score that orders the review queue is the weighted mean of the signals that are available:

$$S = \frac{\sum_{j \in \text{avail}} \alpha_j s_j}{\sum_{j \in \text{avail}} \alpha_j} \tag{4.6}$$

with weights α = 0.4 for entropy, 0.4 for flip disagreement and 0.2 for mask instability, renormalized over the available signals. Detection instability is displayed but not included in S. None of these signals is an ensemble or a Bayesian estimate; all come from one network.

4.4.4 Review Rule

A region r is sent for review when any of the conditions of Table 4.1 holds:

$$\text{review}(r) = [H \geq \tau_H] \vee [JS \geq \tau_{\text{JS}}] \vee [u_{\text{mask}} \geq \tau_M] \vee [d < \tau_D] \vee \text{num} \tag{4.7}$$

where d is the detection score, $\tau_{\text{H}} = 0.397$, $\tau_{\text{JS}} = 0.076$, $\tau_{\text{M}} = 0.30$, $\tau_{\text{D}} = 0.50$, and "num" is true when the numbering check reports a problem or the tooth has no number. The entropy threshold maximizes Youden's index [35] on the calibration crops,

$$J = \text{TPR} - \text{FPR} \tag{4.8}$$

where a positive is a classifier error, TPR is the fraction of errors flagged and FPR the fraction of correct crops flagged.

4.4.5 Evaluation Measures

Precision, recall and F1 are computed from true positives (TP), false positives (FP) and false negatives (FN); macro-F1 is the unweighted mean of the per-class F1 values:

$$P = \frac{\text{TP}}{\text{TP+FP}}, \quad R = \frac{\text{TP}}{\text{TP+FN}}, \quad \text{F1} = \frac{2PR}{P+R}, \quad \text{F1}_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} \text{F1}_k \tag{4.9}$$

Detection and segmentation use COCO-style average precision [27]: AP50 at an IoU threshold of 0.5, and mAP averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05. Calibration is measured with the expected calibration error over 15 equal-width confidence bins [33] and with the multi-class Brier score [34]:

$$\text{ECE} = \sum_{m=1}^{15} \frac{|B_{\text{m}}|}{n} \left|\text{acc}(B_{\text{m}}) - \text{conf}(B_{\text{m}})\right|, \quad \text{BS} = \frac{1}{n}\sum_i \sum_k (p_{ik} - y_{ik})^2 \tag{4.10}$$

Temperature scaling [32] divides the logits by a scalar T fitted by negative log-likelihood on the calibration split (grid from 0.5 to 5.0 in steps of 0.05):

$$p_k = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)} \tag{4.11}$$

For an abstention rule [31], coverage is the fraction of crops that receive a class, and risk is the error rate among them:

$$\text{cov} = \frac{|\text{accepted}|}{n}, \quad \text{risk} = \frac{|\text{wrong} \cap \text{accepted}|}{|\text{accepted}|} \tag{4.12}$$

Finally, the ability of a signal to rank errors is summarized by the AUROC, the probability that a randomly chosen misclassified crop receives a higher score than a randomly chosen correct one, with ties counted as one half.

CHAPTER 5
PROJECT IMPLEMENTATION

5.1 OVERVIEW OF PROJECT MODULES

The implementation has four parts: a Python library (*src/dentexmodel*) that holds the data handling, models, uncertainty, reporting and evaluation code; a FastAPI back end (*backend*) that exposes this library as a local web service; a React front end (*frontend*); and a set of command-line scripts for data processing, training, evaluation and auditing. Table 5.1 maps the main modules to their location in the code. The library can be used without the web application, which is how the training and evaluation scripts use it.

Table 5.1: Main Modules and Their Location in the Code

Module	**Location**	**Responsibility**
Data	src/dentexmodel/data	Discovery, manifest, validation, pixel-hash splits, leakage checks, FDI codes, crop preprocessing
Training	src/dentexmodel/training; scripts/train_model.py	Training configuration, trainers, background jobs, provenance records
Inference	src/dentexmodel/inference	Model registry, torchvision models, tooth-number assignment, Grad-CAM, overlays
Uncertainty	src/dentexmodel/uncertainty	Entropy, flip disagreement, instability, abstention, temperature scaling, quality flags
Reporting	src/dentexmodel/reporting	Evidence bundle, template and LLM drafting, validator, Hindi and Marathi templates, final result
Evaluation	src/dentexmodel/evaluation; scripts/evaluate_*.py	Metrics and the hash-chained test-set ledger
Active learning	src/dentexmodel/active_learning	Random, entropy, disagreement and combined selection
Annotation and review	src/dentexmodel/annotation	Region reviews with revisions and approvals; Label Studio connector
Back end	backend (api, services, database, security.py)	REST API, ten-step pipeline service, SQLite storage, token, loopback and rate-limit checks
Front end	frontend/src (pages, components)	Home, X-ray Analysis, Reports, Review Findings, About and Limitations, Settings, research pages
Tests and audits	tests; scripts/audit_*.py	Automated test suite and live audit harnesses

5.2 DATA COLLECTION AND PREPROCESSING

5.2.1 Data Collection

No new images were collected. The project uses the public DENTEX challenge data [14], distributed through the Zenodo open-access repository and released under the CC BY-NC-SA 4.0 licence, whose terms apply to any reuse. The images are de-identified research data from three institutions; according to the dataset description they come from patients aged 12 and above. The data were stored locally in *Data/training_data* (3,606 files, 14.7 GB) and *Data/validation_data* (51 files).

5.2.2 Dataset Audit and Splitting

Before any training, an audit script identified every radiograph by the SHA-256 hash of its pixels. This revealed the overlap described in Section 3.6: identical radiographs appear in several sub-sets under different names. Splitting by file would therefore have placed copies of the same radiograph in training and test. Instead, all copies form one group, and groups are assigned with seed 42 to train, validation, calibration and test in proportions 0.70/0.10/0.10/0.10. The split file is append-only, so a radiograph never changes split once assigned. Figure 5.1 summarizes the protocol, including the leakage checks and the gate around the test split.

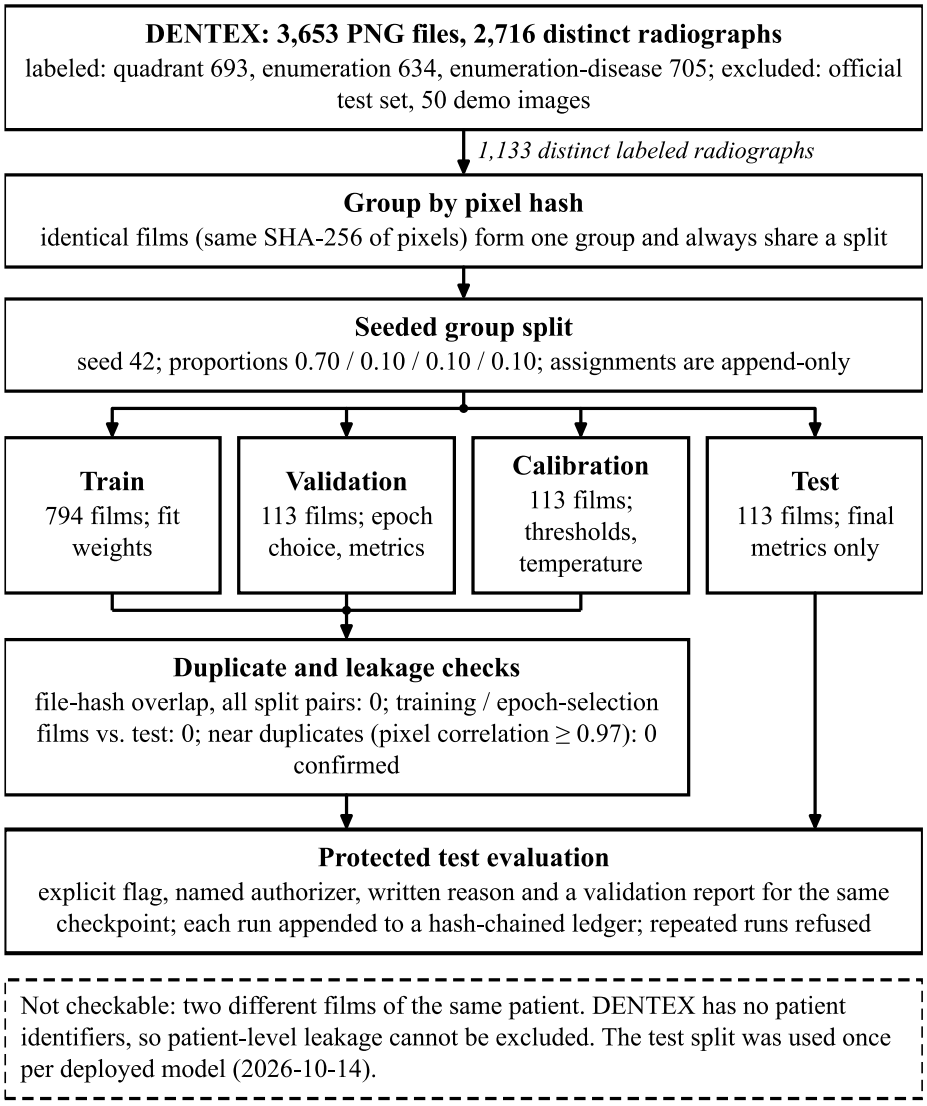


Figure 5.1: Dataset splitting and integrity protocol: pixel-hash grouping, seeded split into four splits, duplicate and leakage checks, and the gated test evaluation.

5.2.3 Image Preprocessing

The uploaded file is never changed. For the detection, numbering and segmentation models the image is converted to an RGB array with values in [0, 1] at full resolution, and torchvision's internal transform resizes it so that the shorter side is 800 pixels and the longer side at most 1,333 pixels. Grayscale contrast statistics are computed only to raise image-quality warnings; they do not alter the model input. For the finding classifier, each tooth box is enlarged by 10 % on each side, cropped, resized to 224 × 224 pixels and normalized with the ImageNet mean and standard deviation [28]. The same crop function is used in training, evaluation and the application, and an end-to-end check confirmed that both paths give byte-identical pixels (Section 7.5).

5.2.4 Labels and Upload Validation

FDI codes are formed from the quadrant and tooth index of the enumeration labels, giving 32 permanent-tooth classes [9]. Finding labels are the four English DENTEX classes. Uploaded files must be PNG or JPEG; both the extension and the file signature are checked, the size limit is 25 MB, and the stored name is derived from the content hash rather than the client's file name. Files that fail any check are refused and nothing is written to disk.

5.3 GENERATIVE AI MODEL SELECTION

The generative component had to turn a list of model findings into readable text in three languages without adding anything. Four options were considered, and Table 5.2 records the decision for each.

Table 5.2: Options Considered for the Generative Component

Option	Reason for the decision	Decision
Image-to-report model such as R2Gen [21]	Needs a large set of images paired with written reports; DENTEX has none	Not possible
Free-text generation by an LLM [4]	Fluent, but can invent teeth, numbers or diagnoses [7]; needs an external service or a large local model	Rejected as the default
Retrieval-augmented generation [8]	Needs a curated, reviewed dental knowledge corpus, which the project does not have	Not used
Template-based data-to-text [19] with optional LLM rewording	Deterministic, works offline, every sentence traceable to a finding; an LLM can reword but cannot bypass the validator	Selected

The selected design keeps generation subordinate to evidence. The template generator is the default and is the only generator that has been used to produce reports. The optional LLM path is implemented for completeness. It sends only de-identified findings to an OpenAI-compatible endpoint, accepts only JSON output, and passes that output through the same validator as the templates, falling back to the template on any violation. Because no endpoint was configured, this path has never been run against a live model; its behaviour is known only from unit tests with a simulated response.

For the perception models, the choice followed from the hardware and from the code base. Faster R-CNN [23], Mask R-CNN [25] and ResNet-50 [26] are available in torchvision with pre-trained weights, fit on a 4 GB laptop GPU, and were already used

by the template notebooks the project started from [16]. A diffusion-based detector like the DENTEX baseline [15] would have needed far more computation.

5.4 MODEL CONFIGURATION AND FINE-TUNING

All four perception models are fine-tuned from general-domain pre-trained weights: the detectors from COCO [27] and the classifier from ImageNet [28]. Table 5.3 gives the configuration recorded in each checkpoint's sidecar file. Every model uses AdamW with weight decay 10^{-4}, a cosine learning-rate schedule and seed 42, and keeps the epoch with the best validation score. Horizontal flips augment the one-class detector and the segmenter, but not the numbering model, because a flip would swap left and right FDI codes. The classifier uses horizontal flips with probability 0.5, multiplicative contrast jitter of ±5 % and the class weights of (4.2).

Table 5.3: Training Configuration of the Four Deployed Checkpoints

Model	Architecture (initialization)	Epochs / batch / lr	Selection on validation
Tooth detector	Faster R-CNN R50-FPN, 1 class (COCO)	12 / 2 / 1e-4	box mAP@[.5:.95]
FDI numbering	Faster R-CNN R50-FPN, 32 classes (COCO)	16 / 2 / 1e-4	box mAP@[.5:.95]
Segmenter	Mask R-CNN R50-FPN, 1 class (COCO)	8 / 1 / 1e-4	box mAP@[.5:.95]
Finding classifier	ResNet-50, 4 classes (ImageNet V2)	15 / 32 / 3e-4	macro-F1 (epoch 5 kept)

Sources: Data/checkpoints/*.json sidecars. Training data: 440 radiographs for the three localization models and 2,520 crops from 476 radiographs for the classifier. The checkpoints are legacy: image-level training membership was not recorded (Section 7.6).

The optional language model is not fine-tuned. If configured, it is called with temperature 0, a JSON response format and a system prompt that tells it to use only the supplied findings, never to say that a finding is present, to mark high-uncertainty findings as needing human review, and never to add diagnoses, treatment advice, teeth, labels or numbers. The endpoint, key and model name are read from environment variables and are never returned by the API.

5.5 PROJECT MODULES

5.5.1 Image Management

The image service lists dataset and uploaded images under identifiers that are unique across sub-sets, serves files, thumbnails and quality summaries, and accepts uploads after the checks of Subsection 5.2.4. Identifiers are never used as file paths, so a request cannot reach files outside the manifest or the upload table.

5.5.2 Analysis Pipeline

The pipeline service runs the ten steps of Figure 4.3 for one image. It applies the detector, assigns FDI numbers (Algorithm Algorithm 2), runs the segmenter, matches masks to boxes, classifies every crop and its flipped copy, repeats detection and segmentation on the flipped image, and stores one record per region with a content-derived identifier built from the image, the model version and the box rounded to 8 pixels. Identical reruns therefore give identical region identifiers, which keeps reviews attached to the right region.

5.5.3 Uncertainty and Review Routing

The uncertainty module computes the signals of Section 4.4 and applies the review rule (4.7). Thresholds are read from the calibration report and are labelled provisional in the interface. The review queue orders images by the combined score S, so that the most uncertain images are reviewed first.

5.5.4 Explainability

For a selected region, the explain endpoint computes a Grad-CAM heat map [37] of the classifier for that crop and returns it at the size of the box. The heat map shows which pixels influenced the classifier's score. It does not show where a finding is, and the interface says so.

5.5.5 Human Review

A reviewer can confirm, correct or reject a region's class, set or clear its tooth number, or mark the image for specialist review or as unusable. Every record stores the model output the reviewer saw. A changed decision creates a new revision, an identical resubmission is refused, and approval requires a second person whose name differs from the reviewer's. Reviewer names are typed in and are not verified. Only approved reviews

of unlabeled-pool or uploaded images can be exported for training; validation, calibration, test and labeled images are excluded and counted.

5.5.6 Report Generation and Localization

The report service builds the evidence bundle, calls the template generator (or, if configured, the LLM path), validates the draft, stores it and renders HTML, TXT or JSON exports in English, Hindi or Marathi (Algorithm Algorithm 3). The Hindi and Marathi views come from fixed sentence templates (src/dentexmodel/reporting/i18n.py); a localization check refuses an export if any identifier, FDI code, score, version or review status differs from the English report. Every view carries the research-draft disclaimer, the limitations and, for Hindi and Marathi, a notice that the translation awaits linguistic review.

5.5.7 Active Learning

The active-learning module scores unlabeled pool images with the uncertainty pipeline and selects images for annotation by random, entropy, disagreement or combined strategies, with a recorded seed and budget. The pool excludes the locked test split and pixel duplicates of labeled images. Signals that cannot be computed, such as ensemble disagreement with a single model, are reported as unavailable rather than filled with zeros.

5.5.8 Model Registry, Training and Evaluation

The registry lists each checkpoint with its sidecar metadata and refuses a checkpoint whose file hash does not match the sidecar. Training jobs run in the background with per-epoch metrics and can be stopped. Evaluation through the API is limited to the validation and calibration splits; the test split can be evaluated only from the command line with an explicit flag, a named authorizer, a written reason and a prior validation report for the same checkpoint, and every such run is appended to a hash-chained ledger.

5.5.9 User Interface

The front end offers a simple path for a reviewer (Home, X-ray Analysis, Reports, Review Findings, About and Limitations, Settings) and separate research pages for uncertainty, models, training, experiments and active learning. The analysis page overlays boxes, tooth numbers, candidate classes and masks on the radiograph, lists every

region and states how many need review. Screenshots of the implemented interface are shown in Section 7.8.

5.6 TOOLS AND TECHNOLOGIES USED

Table 5.4 lists the tools used and what each was used for. The list is limited to tools that are part of the delivered system or were used to produce its results.

Table 5.4: Tools and Technologies Used

Tool	**Purpose in the project**
Python 3.13	Library, back end, training, evaluation and audit scripts
PyTorch 2.6 and torchvision 0.21	Model definitions (Faster R-CNN, Mask R-CNN, ResNet-50), pre-trained weights, training and inference on CUDA 12.4
NumPy and Pillow	Image decoding, arrays, masks and crops
FastAPI, Uvicorn and Pydantic	REST API, request validation and error responses
SQLite	Storage of images, analyses, regions, reviews, reports and jobs
React 18, React Router, Recharts, lucide-react	Single-page user interface, charts and icons
Vite 5 and Tailwind CSS 3	Front-end development server with API proxy, build and styling
pytest and unittest	Automated test suite (590 tests passed on 2026-10-26)
Playwright with Microsoft Edge	Browser audits of the user interface
Git	Version control of code and reports
Windows PowerShell scripts	Setup, start-up, health check and the audit suite on Windows

5.7 ALGORITHM DETAILS

Three algorithms capture the core of the implementation. Algorithm 1 is the per-image analysis and review routing; Algorithm 2 assigns FDI numbers to detector boxes; Algorithm 3 drafts and validates a report.

Algorithm 1: Per-image analysis and review routing (as implemented)

Input: radiograph I (RGB, values in [0, 1]); thresholds $\tau_H = 0.397$, $\tau_{JS} = 0.076$, $\tau_M = 0.30$, $\tau_D = 0.50$

1: $D \leftarrow$ tooth detector(I), keeping boxes with score ≥ 0.3
2: assign FDI codes to D with Algorithm 2 and run the consistency check
3: $M \leftarrow$ segmenter(I); match M to D one to one at IoU ≥ 0.5
4: repeat detection and segmentation on the flipped image to obtain instability signals
5: **for** each box $d \in D$ **do**
    6: $\mathbf{p} \leftarrow$ classifier(crop(d)); $\mathbf{q} \leftarrow$ classifier(flip(crop(d)))
    7: compute $H(\mathbf{p})$, JS($\mathbf{p}$, $\mathbf{q}$), u_{mask} and detection instability
    8: flags $\leftarrow$ conditions of the review rule that hold, plus missing mask and "not found on flipped image"
    9: status $\leftarrow$ abstained if $H \geq \tau_H$ (top class still shown); review $\leftarrow$ flags not empty or number unassigned
10: $S \leftarrow$ weighted mean of the available image-level signals; store region records
11: place the image in the review queue ordered by S

Algorithm 2: FDI number assignment

Input: detector boxes D with scores; raw numbering detections N (32 classes)

1: drop numbering boxes with score < 0.3; apply class-agnostic NMS at IoU 0.5; keep the best box per FDI code
2: **for** each $d \in D$ in descending score order **do**
    3: $n^* \leftarrow$ unused numbering box with the highest IoU(d, n)
    4: **if** IoU(d, n^*) ≥ 0.5 **then** code(d) $\leftarrow$ code(n^*); mark n^* used **else** code(d) $\leftarrow$ unassigned
5: flag codes that are duplicated, have confidence < 0.5, or lie on the wrong side or arch for their quadrant (image-left is patient-right)

Algorithm 3: Evidence-grounded report drafting and validation

Input: stored analysis A; mode $\in$ {template, llm, auto}

1: $E \leftarrow$ evidence bundle: one finding per region of A with id Fk, FDI code, label, raw score, uncertainty, review status

2: **if** mode = llm, or mode = auto and an endpoint is configured, **then**

3: send de-identified findings; $R \leftarrow$ parsed JSON reply; **if** not valid JSON **then** $R \leftarrow$ template(E) and record the rejection

4: **else** $R \leftarrow$ template(E)

5: $V \leftarrow$ validate(R, E): required fields, matching values, valid FDI codes, no unsupported teeth, numbers, finding names or clinical phrases

6: **if** V failed and R came from the LLM **then** $R \leftarrow$ template(E) and record the reasons

7: store R with status "unconfirmed research draft"; render English, Hindi or Marathi on request

8: refuse an export whose localized values differ from R; set status "reviewed" only when a named reviewer signs

CHAPTER 6
SOFTWARE TESTING

Testing was done at four levels: an automated unit and integration suite; live harnesses that call the running API with real checkpoints; browser audits that drive the real interface; and a set of acceptance tests that follow a user through the whole workflow. This chapter reports software tests only. The accuracy of the models is a separate question and is evaluated in Chapter 7; a passing software test says nothing about clinical validity.

6.1 TEST CASES

Table 6.1 lists the 20 acceptance tests run on 2026-10-14 with the real FastAPI back end, the real Vite front end and real checkpoints on real radiographs. No mocked inference was used. Nineteen tests passed and one (AT-17) was partial, because three state-changing buttons were deliberately not clicked and were tested through the API instead.

Table 6.1: Acceptance Test Cases and Observed Outcomes (2026-10-14)

ID	Feature	Input or scenario	Expected outcome	Observed outcome	Status
AT-01	Back end	Start FastAPI	Server starts	Healthy 4.1 s after launch	Pass
AT-02	Front end	Start Vite	UI on port 5173 with API proxy	HTTP 200; proxy verified	Pass
AT-03	Health	GET /api/health	Database ok, 4 checkpoints	Ok; health check script exit 0	Pass
AT-04	Upload	Valid PNG, JPEG, Unicode file name	Stored under hash id; bytes unchanged	Accepted; duplicate detected; download byte-identical	Pass
AT-05	Upload	Corrupt, truncated, wrong type, empty, oversized files	4xx; nothing stored	All refused with 422; nothing left on disk	Pass
AT-06	Detection	Two test-split films and one upload	Real boxes, valid geometry	31, 28 and 33 detections on GPU in about 0.4 s; identical on repeat	Pass
AT-07	Classification	Detected regions	Softmax per crop; abstention	Four scores summing to 1; high-entropy regions abstained	Pass
AT-08	Segmentation	Same images	Valid masks	32, 29 and 32 instances inside their boxes	Pass
AT-09	Overlay	Compare UI with API output	Identical	All 32 rows and boxes equal; position within 0.3 px	Pass
AT-10	Numbering missing	Remove numbering checkpoint	No invented numbers	0 of 32 regions numbered; all flagged; step unavailable	Pass
AT-11	Uncertainty	Compute for an image	Real signals; unavailable ≠ 0	Entropy 1.0 for uniform and 0.0 for one-hot input; thresholds labelled provisional	Pass
AT-12	English report	Generate from analysis	Research-draft banner	Generated; banner shown	Pass

ID	Feature	Input or scenario	Expected outcome	Observed outcome	Status
AT-13	Hindi, Marathi	Switch language	Identical structured values; no new inference	58 report checks identical; no inference request	Pass
AT-14	Export	HTML, TXT, JSON per language	Well-formed files	9 exports well-formed and escaped	Pass
AT-15	Persistence	Reload; second server process	Same report	Reopened with same content and list	Pass
AT-16	Model registry	List models	Version, hash, evaluation links	4 models; sidecar hash equals file hash	Pass
AT-17	UI controls	Click every control	No silent buttons	56 controls work or state why not; 3 state-changing buttons not clicked	Partial
AT-18	Logs	Access log and console	No unexpected errors	1,650 log lines, no HTTP 500, no traceback	Pass
AT-19	Unit tests	Run test suite	All pass	180 tests OK	Pass
AT-20	Data safety	Hash checkpoints and data	Nothing deleted or overwritten	All 8 checkpoint files and the data folders identical	Pass

Source: docs/FINAL_ACCEPTANCE_REPORT.md and docs/audit_evidence/. The abstention display described in AT-07 and AT-12 changed on 2026-10-16: abstained regions now also show their highest-scoring class with a review label.

Table 6.2 summarizes the larger test campaign. The first audit pass found one cosmetic failure (the health endpoint reported its version as unknown); it was fixed in a second pass the same day, which also added authentication, loopback restriction and rate limiting. On 2026-10-26, for this report, the automated suite was run again on the current code.

Table 6.2: Summary of Software Test Results

Test campaign	Date	Result
Acceptance tests (Table Table 6.1)	2026-10-14	19 pass, 1 partial, 0 fail
All harnesses, first pass (unit, live API, per-model, reports, browser, review)	2026-10-14	466 pass, 1 fail (cosmetic), 0 blocked
All harnesses, second pass	2026-10-14	550 pass, 0 fail (including 205 unit tests)
Security checks (token, loopback, rate limits)	2026-10-14	23 pass
Functionality audit (60 features)	2026-10-14	42 pass, 1 fail, 5 blocked, 7 partial, 5 not implemented; 15 defects fixed
Full test suite after the last training run	2026-10-16	309 pass, 0 fail
Full test suite (pytest), current code	2026-10-26	590 pass, 0 fail (and 21 subtests) in 197 s

Sources: docs/FINAL_ACCEPTANCE_REPORT.md, docs/REMAINING_ISSUES.md, docs/FUNCTIONALITY_AUDIT.md, reports/training_summary.md; the 2026-10-26 run was executed for this report. Blocked items (live LLM, Label Studio, TensorBoard, native linguistic review) are not counted as passed.

6.2 GENERATIVE AI OUTPUT TESTING

The report generator was tested as a component that must be correct, not merely fluent. The campaign of 2026-10-14 ran 61 checks against the live API; all 61 passed. A further 21 report rows in the live-API harness and seven browser checks also passed. Table 6.3 shows a representative selection. The checks cover four questions: is the report built only from the analysis, do the three languages carry exactly the same values, does the validator reject bad evidence and unsupported text, and are failures reported truthfully.

Table 6.3: Selected Report-Generation Test Cases (2026-10-14)

ID	Check	Expected	Observed	Status
RPT-A02	Generate English report from a real 33-region analysis	Unconfirmed draft with model versions	Created with status unconfirmed research draft	Pass
RPT-A03	Regenerate from the same analysis	Identical findings	New id; findings identical	Pass
RPT-B02	Unsupported language codes (xx, fr, HI, path-like)	422, no English fallback	All refused with 422	Pass
RPT-C01	Hindi and Marathi values vs English	Exactly equal	33 of 33 findings equal in each language	Pass
RPT-C03	Every decimal of the English text in the translation	None missing	None missing	Pass
RPT-C07/C08	Treatment wording in any language	None	None found	Pass
RPT-C09	Sentences mentioning diagnosis or presence	All negations or limitations	16 sentences, all negated	Pass
RPT-E04	Evidence with invalid FDI code 99	Report flagged invalid	Validation failed; not exportable as clean	Pass
RPT-E05	Evidence with label “Healthy”	Report flagged invalid	Validation failed	Pass
RPT-E06	Probability 1.7	Report flagged invalid	Validation failed	Pass
RPT-E07	Injected “confirmed diagnosis” text	Truthful refusal or safe, flagged output	HTTP 200; output met the harness’s safety criterion	Pass
RPT-E08	LLM mode without configuration	Truthful refusal	502 “LLM is not configured” with a hint	Pass
RPT-F01	Sign-off with empty reviewer name	Refused	422; status stays unconfirmed	Pass
RPT-I02	Translation review flag	Shown as not reviewed	Flag false for Hindi and Marathi	Pass
REP-031	Reviewer name containing a script tag	Escaped in HTML export	No raw script tag	Pass

Source: docs/MULTILINGUAL_REPORT_TEST_REPORT.md (scripts/audit_reports.py and scripts/audit_live_api.py).

The LLM path could not be tested against a real model, because no endpoint was configured. Its logic is covered by unit tests that replace the network call with a simulated reply. A valid JSON reply is accepted and marked as generated by the LLM, and the outgoing request is checked to contain no image identifier. A reply whose summary says “Tooth 36 needs a root canal.” is rejected and replaced by the template draft, because

tooth 36 is not in the evidence and “root canal” is a forbidden phrase. A reply that is not JSON is also replaced by the template. These tests show that the guard works as designed; they do not show how often a real model would trip it.

The campaign also found and fixed three defects in the validator. Evidence with an impossible probability (1.7) or a label no model can produce (“Healthy”) had been accepted, because the validator compared the report with its evidence but did not check the evidence itself. A report about “Deep Caries” was wrongly flagged for mentioning “caries”. And the first fix for the label check broke every real report, because the configuration spells the class “DeepCaries” while the model outputs “Deep Caries”; the live run caught this before release. Each defect now has a regression test.

Two qualities could not be tested by software. The linguistic quality of the Hindi and Marathi wording needs a native-speaking dental professional, and no such review has taken place; the interface and every export say so. Whether the drafts are useful to a clinician also remains unmeasured.

6.3 PERFORMANCE EVALUATION

Table 6.4 gives the inference time of each stage on the RTX 3050 Laptop GPU for 10 panoramic images of up to 3,076 × 1,536 pixels, measured after two warm-up runs with no failures. Detection takes about 0.1 s per image and segmentation about 0.3 s, with peak GPU memory up to 1.65 GB. The detection-plus-numbering harness runs the detector twice and so overstates the cost in the application.

Table 6.4: Inference Benchmark (RTX 3050 Laptop GPU, 10 Images)

Stage	Mean (s)	Median (s)	p95 (s)	Images/s	Peak memory (MB)
Detection	0.110	0.099	0.153	9.08	613
Segmentation	0.285	0.298	0.339	3.51	1,653
Detection + numbering	0.334	0.332	0.367	2.99	985
Detection + classification	0.252	0.246	0.291	3.96	1,039

Source: Data/reports/benchmark_inference.json (2026-10-14).

In the archived interface audit, the application reported 1.92 s for the complete analysis of one image, including the flipped passes, and the server was healthy 4.1 s after a clean start. Without a GPU the same models run on the CPU, at about 7 to 23 s per

prediction and 11 to 53 s per full analysis on the same laptop. These timings show that the system is interactive on modest hardware; they do not bear on its accuracy.

CHAPTER 7
RESULTS AND DISCUSSION

This chapter reports what the project produced and how well it works. Model results come from evaluation reports stored in the project and are copied without rounding beyond three decimals. Each table names its source. Validation numbers are optimistic, because validation also selected each model's training epoch. The internal test split was evaluated once for each of the four deployed checkpoints on 2026-10-14, after model selection. All models were trained with one seed on one dataset, so there are no confidence intervals over seeds for the deployed models.

7.1 PROJECT OUTCOMES

The project delivered the following:

- an audited DENTEX data pipeline with pixel-hash grouped splits, leakage checks and a hash-chained ledger of test-split use;
- four trained and registered models (tooth detector, FDI numbering model, tooth instance segmenter and four-class finding classifier), each with a sidecar recording its configuration, data fingerprint and hash;
- a ten-step analysis pipeline that links the models region by region and attaches uncertainty signals and review flags to every region;
- an evidence-grounded report generator with English, Hindi and Marathi output, an automatic validator and named-reviewer sign-off;
- a local web application (FastAPI and React) with analysis, review, reporting, explanation, active-learning and model-management functions;
- a test campaign with 20 acceptance tests and an automated suite of 590 passing tests; and
- an evaluation that includes the unfavourable and negative results reported in this chapter, together with a 10-page IEEE-format draft paper (Appendix A).

7.2 RESULT ANALYSIS

Figure 7.1 maps each task and study to its metrics and to the splits on which it was evaluated. It separates two settings that are easily confused: classification on ground-truth tooth crops, which measures the classifier alone, and end-to-end classification on detector crops, which measures what the application actually does.

Task	Metrics	Val.	Cal.	Test
MODELS				
Tooth detection	mAP, AP50, P / R @ 0.3	●		●
Segmentation	box / mask AP50, IoU, Dice	●		●
FDI numbering	AP50, coverage, accuracy	●		●
Classification, GT crops	P / R / F1, macro-F1, AUC	●		●
SYSTEM				
End-to-end, detector crops	matched / missed, macro-F1	●		
Calibration, uncertainty	ECE, Brier score, AUROC		●	●
Abstention rules	coverage, risk, macro-F1	●	●	
STUDIES				
Active learning	macro-F1 per label budget	●		
Classifier variants	paired macro-F1, 5 seeds	●		
Leakage check	overlaps, all split pairs	●	●	●
Inference benchmark	latency, peak memory	10 images		

Figure 7.1: Experimental evaluation framework: each task or study, its metrics and the splits on which it was evaluated.

7.2.1 Tooth Detection and Segmentation

On the 63 test radiographs (1,790 teeth) the detector reaches AP50 0.966 and recall 0.974 at the operating point (score ≥ 0.3, IoU ≥ 0.5), as Table 7.1 shows. The segmenter's mask AP50 is 0.974, and matched instances have a mean mask IoU of 0.780 and a mean Dice coefficient of 0.873. The much lower mAP@[.5:.95] (0.515 for detection) means that the boxes lie close to the annotations but are not tight. That difference looks harmless here, but it matters once the classifier receives detector crops (Section 7.5).

Table 7.1: Tooth Detection and Instance Segmentation

Metric	Det. val	Det. test	Seg. val	Seg. test
Radiographs / GT teeth	69 / 2,000	63 / 1,790	69 / 2,000	63 / 1,790
Box mAP@[.5:.95]	0.560	0.515	0.569	0.530
Box AP50	0.978	0.966	0.981	0.974
Mask AP50	n/a	n/a	0.980	0.974
Mean mask IoU / Dice (matched)	n/a	n/a	n.r.	0.780 / 0.873
TP / FP / FN at score 0.3	1968/78/32	1743/73/47	1972/70/28	1754/69/36
Precision at score 0.3	0.962	0.960	0.966	0.962
Recall at score 0.3	0.984	0.974	0.986	0.980

Sources: Data/reports/eval_dentex_teeth_detection_v1_{val,test}.json and eval_dentex_teeth_segmentation_v1_{val,test}.json. n.r. = not recorded in the validation report. Mask IoU on a 4× subsampled grid; 1,753 matched test instances.

7.2.2 FDI Tooth Numbering

On test, 96.9 % of ground-truth teeth are matched by a numbered prediction and 96.8 % of those receive the exact FDI code (Table 7.2). Counting missed teeth as errors gives a strict accuracy of 93.8 %. Quadrant accuracy is 0.999, so almost all numbering errors put the wrong index in the right quadrant. Test accuracy is higher than validation accuracy (0.968 against 0.936); with 63 and 69 radiographs a gap of this size is within small-sample variation, and we do not read anything into it.

Table 7.2: FDI Tooth Numbering

Metric	Validation	Test
Radiographs / GT teeth	69 / 2,000	63 / 1,790
Box mAP@[.5:.95] / AP50 (32 classes)	0.524 / 0.928	0.507 / 0.952
Precision / recall at score 0.3 (any class)	0.872 / 0.992	0.882 / 0.983
Coverage (matched / GT)	0.979 (1,957)	0.969 (1,734)
Exact accuracy among matched teeth	0.936	0.968
Quadrant accuracy among matched teeth	0.997	0.999
Strict accuracy over all GT teeth	0.916	0.938
Lowest per-tooth exact accuracy	36: 0.825; 47: 0.862	47: 0.912; 27: 0.917

Source: Data/reports/eval_dentex_fdi_numbering_v1_{val,test}.json. Third molars have the fewest ground-truth teeth (34 to 50 per split), so their per-tooth accuracies are the least stable.

7.2.3 Finding Classification on Ground-Truth Crops

On ground-truth test crops the classifier reaches accuracy 0.785 and macro-F1 0.646 (Table 7.3). Impacted teeth are recognized reliably (F1 0.926, AUC 0.990) and Caries reasonably well (F1 0.842). Deep Caries (F1 0.453) and Periapical Lesion (F1 0.364, recall 0.308) are poor. The confusion matrices in Table 7.4 locate the main errors.

Caries and Deep Caries are confused in both directions (30 and 22 test crops), which is not surprising because the two classes differ in depth rather than in kind. Nine of the 13 test Periapical Lesions are labelled as some form of caries. The macro ROC AUC of 0.910 shows that the scores rank the classes better than the argmax decision suggests, which is what makes the uncertainty signals of Section 7.4 useful.

Table 7.3: Finding Classification on Ground-Truth Crops

Class	Split	P	R	F1	n	Spec.	AUC
Caries	val	0.885	0.863	0.874	205		
	test	0.866	0.820	0.842	228	0.770	0.895
Deep Caries	val	0.500	0.467	0.483	45		
	test	0.414	0.500	0.453	48	0.889	0.834
Impacted	val	0.919	0.944	0.932	72		
	test	0.887	0.969	0.926	65	0.972	0.990
Periapical Lesion	val	0.467	0.778	0.583	9		
	test	0.444	0.308	0.364	13	0.985	0.921
Macro / accuracy	val	0.693	0.763	0.718	331	acc. 0.825	
	test	0.653	0.649	0.646	354	acc. 0.785	0.910

Source: Data/reports/eval_dentex_findings_classification_v1_{val,test}.json (65 and 73 radiographs). Specificity and one-vs-rest AUC were computed for the test split only. Validation macro-F1 was also the epoch-selection criterion.

Table 7.4: Confusion Matrices of the Finding Classifier (Rows: True Class; Validation / Test)

True \ predicted	Caries	Deep Caries	Impacted	Periapical
Caries	177 / 187	18 / 30	5 / 6	5 / 5
Deep Caries	20 / 22	21 / 24	1 / 2	3 / 0
Impacted	2 / 1	2 / 1	68 / 63	0 / 0
Periapical Lesion	1 / 6	1 / 3	0 / 0	7 / 4

Counts of ground-truth crops. Source: same reports as Table 7.3.

7.3 GENERATED OUTPUT ANALYSIS

Table 7.5 reproduces the beginning of a real report exported by the application during the audit of 2026-10-14. It was generated by the template path from a full analysis of an uploaded radiograph with 33 detected regions. The text is quoted exactly, including the British spelling used in the templates.

Table 7.5: Excerpt of a Generated English Report (Exported 2026-10-14)

Report text (verbatim)
[UNCONFIRMED RESEARCH DRAFT] Research draft generated from model outputs. Not a clinical diagnosis and not clinically validated. Requires review by a qualified professional.
Summary: The detection model (dentex_teeth_detection_v1) located 33 tooth region(s). Segmentation masks come from the segmentation model (dentex_teeth_segmentation_v1) and are matched to regions by overlap. Tooth numbers (FDI) come from the numbering model (dentex_fdi_numbering_v1); regions it did not match stay unnumbered. The classification model (dentex_findings_classification_v1) gave each region its closest class among four trained findings. It has no healthy class, so a class name here is not a statement that the finding is present. Of these, 32 have a tooth number from the numbering model, 32 have a matched segmentation mask, and 22 are flagged for manual review.
F1: tooth 48 - closest trained class 'Impacted' (score 0.98, normalised uncertainty 0.08); not evidence that the finding is present. Review status: unreviewed.
F2: tooth 18 - closest trained class 'Caries' (score 0.93, normalised uncertainty 0.19); not evidence that the finding is present. Review status: unreviewed.
F3: an unnumbered region - no reliable candidate class (normalised uncertainty 0.61); manual review required. Review status: unreviewed.

Source: docs/audit_evidence/report_en.txt (report RPT-5969557C37, generated by: template). Since 2026-10-16, an abstained region such as F3 is described with its highest-scoring class and a statement that it needs human review because of high classifier uncertainty.

Three properties of the generated text are worth pointing out. First, every number in it comes from the evidence: the region count, the counts of numbered, masked and flagged regions, and each score and uncertainty value. The validator checks this; a decimal number that is not in the evidence would make the report invalid. Second, every statement about findings is hedged in the same way. The classifier's output is called the "closest trained class" and is followed by "not evidence that the finding is present". The audit counted 16 sentences that mention diagnosis or presence in an English report, and all 16 were negations or limitations. Third, the report names the model version behind each part of the analysis, so that a reader can trace a statement back to the checkpoint that produced it.

The cost of this design is fluency. The text is repetitive, and a clinician reading 33 findings would probably prefer a shorter summary grouped by quadrant or by finding. That is the kind of rewording the optional language-model path was meant to provide, under the validator's control. Because no endpoint was configured, we have no evidence on whether a real model would produce better text without being rejected, and no claim is made about it.

The Hindi and Marathi views (Figure 7.5) contain the same identifiers, FDI codes, scores and review statuses as the English report. In the audit, all 33 findings were identical in each language, every decimal value of the English text appeared verbatim, and no

untranslated English prose or treatment wording was found. Their wording has not been reviewed by a native-speaking dental professional, and every Hindi and Marathi export carries a notice saying so.

7.4 PERFORMANCE METRICS

7.4.1 Summary of Model Metrics

Table 7.6 collects the headline test metrics from Section 7.2 in one place, with the weakest class for each task, so that the strengths and the weaknesses can be read side by side.

Table 7.6: Summary of Test Metrics

Task	Main test metric	Weakest point
Tooth detection	AP50 0.966; recall 0.974; precision 0.960	boxes not tight: mAP@[.5:.95] 0.515
Instance segmentation	mask AP50 0.974; mean Dice 0.873	mean mask IoU 0.780
FDI numbering	exact accuracy 0.968 among matched teeth; strict 0.938	teeth 47 and 27 (0.912 and 0.917)
Finding classification (GT crops)	accuracy 0.785; macro-F1 0.646; macro AUC 0.910	Periapical Lesion recall 0.308; Deep Caries F1 0.453
Error capture by entropy flag	86.8 % of errors inside the flagged 54.2 % of crops	13.2 % of errors missed; over half of all crops flagged

Test split: 63 radiographs for detection, segmentation and numbering; 73 radiographs (354 crops) for classification.

7.4.2 Uncertainty and Calibration

Thresholds and the temperature were fitted on the calibration split and then applied once to the test crops (Table 7.7). The entropy flag marks 54.2 % of test crops. Their error rate is 34.4 %, against 6.2 % for unflagged crops, and they contain 86.8 % of all errors (66 of 76). The flag therefore catches most errors, but it misses 13.2 % of them and sends more than half of the regions to review. The signals carry real information about errors (AUROC 0.768 for entropy and 0.745 for flip disagreement), although the simpler 1 − max-probability score ranked test errors slightly better (0.790).

Table 7.7: Uncertainty and Calibration of the Finding Classifier

Measure	Calibration	Test
Crops / accuracy	324 / 0.830	354 / 0.785
ECE (15 bins) / Brier score, raw scores	0.071 / 0.277	0.048 / 0.291
ECE / Brier after temperature T = 0.95	0.076 / 0.276	0.036 / 0.291
AUROC for errors: entropy	0.721	0.768
AUROC for errors: flip disagreement	0.709	0.745
AUROC for errors: 1 − max probability	0.713	0.790
Crops flagged by entropy ≥ 0.397	threshold fitted	54.2 %
Error rate, flagged / unflagged	n/a	0.344 / 0.062
Share of all errors inside the flagged set	n/a	0.868
Flip-unstable (≥ 0.076): share; error rate vs. stable	90th percentile	9.3 %; 0.485 vs. 0.187

Source: Data/reports/uncertainty_evaluation.json (2026-10-14; test-ledger entry 5). Temperature scaling is not applied in the application.

Temperature scaling gave a mixed result. The fitted temperature (0.95) lowered test ECE from 0.048 to 0.036, but it raised calibration ECE from 0.071 to 0.076 on the very split it was fitted on and left the Brier score almost unchanged. Because the effect is small and inconsistent between splits, the application keeps raw scores and labels them as uncalibrated.

7.4.3 Flip Instability of the Detector and Segmenter

A separate check on 30 validation radiographs, run through the live API with the real models, asked whether flip instability flags poor masks or false detections (Table 7.8). Mask instability ranked masks with IoU below 0.8 only weakly, and detection instability ranked false positives with AUROC 0.644; with only 27 false positives among 878 detections, that second estimate rests on few events. Neither signal should be presented as a validated error detector, and the interface labels both as weak.

Table 7.8: Flip-Instability Validation (30 Validation Radiographs)

Signal	Regions	Target event	AUROC
Mask instability	851	mask IoU with ground truth < 0.8 (51.8 %)	0.593
Detection instability	878	false-positive box (27; 3.1 %)	0.644

Source: Data/reports/instability_validation.json (2026-10-14). Error rate above vs. at or below the median mask signal: 0.598 vs. 0.439.

7.4.4 Abstention Rules

An offline experiment compared 33 abstention rules (entropy, maximum-probability and top-2-margin thresholds) on detector crops (Table 7.9). Rules were selected on calibration and confirmed on validation; the test split was never read. Only

"abstain if max $p < 0.75$" passed the calibration criteria. On validation it raised coverage from 0.520 to 0.638 at almost the same risk (bootstrap 95 % confidence interval of the risk difference from −0.018 to +0.026), but it failed the minority-class criterion, because Periapical Lesion precision fell from 0.50 to 0.43. The production entropy rule was therefore kept.

Table 7.9: Abstention Rules on Detector Crops

Rule / split	Coverage	Risk	Macro-F1 (all)	Periapical P / R
Entropy ≥ 0.397, calibration	0.506	0.093	0.570	0.80 / 0.50
max p < 0.75, calibration	0.623	0.096	0.620	0.75 / 0.56
Entropy ≥ 0.397, validation	0.520	0.119	0.518	0.50 / 0.43
max p < 0.75, validation	0.638	0.121	0.571	0.43 / 0.43

Source: docs/audit_evidence/ml_improvement/abstention/report.md (2026-10-15). Calibration 318 and validation 323 matched findings. "All" counts abstained and missing crops as not recalled.

7.5 ACCURACY, RELEVANCE AND QUALITY EVALUATION

7.5.1 End-to-End Accuracy on Detector Crops

The classifier results of Section 7.2 assume perfect boxes. A dedicated study ran the production path (detector, crop, classifier) on the 65 validation radiographs of the findings sub-set, with 331 findings, without touching calibration or test data (Table 7.10). It first confirmed that production and evaluation preprocessing give byte-identical pixels and identical predictions on all 331 crops, which rules out a preprocessing mismatch. On the 24 radiographs that also have a complete tooth annotation, the detector found 716 of 719 teeth at IoU 0.5 but only 582 at IoU 0.75. Detector boxes matched 323 of the 331 findings; of the 8 unmatched findings, 4 were genuine detector misses and 4 were second findings on a tooth that one box could match only once.

Table 7.10: End-to-End Validation on Detector Crops (65 Radiographs, 331 Findings)

Measure	GT boxes	Detector boxes
Detection on 24 fully annotated films: P / R / F1 at IoU 0.5	n/a	0.977 / 0.996 / 0.986
Same at IoU 0.75	n/a	0.794 / 0.809 / 0.802
Same 323 matched teeth: accuracy	0.830	0.811
Same 323 teeth: macro-F1	0.717	0.649
Same 323 teeth: macro recall	0.777	0.688
All 331 findings (miss counted as wrong): accuracy	0.825	0.792
All 331 findings: macro-F1	0.718	0.634
All 331: Deep Caries recall (n = 45)	0.467	0.333
All 331: Periapical Lesion recall (n = 9)	0.778	0.444
Unlabeled detector boxes given a finding class	n/a	1,580 of 1,903

Source: Data/reports/end_to_end_validation_evaluation.md and .json (2026-10-15). Validation also selected the epochs, so all numbers are optimistic.

Accuracy fell from 0.825 to 0.792, a net loss of 11 findings: 20 findings became wrong with detector boxes, 14 became right and 5 correct findings were lost to non-matching. About 10 % of predictions changed even when the detector box overlapped the annotation with IoU between 0.8 and 1.0, so the classifier is sensitive to small shifts of the crop and not only to badly placed boxes. The loss falls mostly on the rare classes: Periapical Lesion recall fell from 7 of 9 to 4 of 9, and its precision on matched teeth from 0.46 to 0.29. Eight of the 331 findings (2.4 %), including 4 of the 9 validation Periapical Lesions, sit on teeth that carry a second finding, and a single-label classifier cannot be right for both.

7.5.2 Relevance of the Output

The largest practical problem lies outside these metrics. Of the 1,903 detector boxes on the 65 radiographs, 1,580 match no annotated finding, yet without a “no finding” class the application names a closest class for each of them. On the 24 fully annotated films, 584 such boxes cover real teeth without a finding label, and the classifier called them Caries 475 times, Periapical Lesion 71 times, Deep Caries 23 times and Impacted 15 times. DENTEX does not certify that unlabeled teeth are healthy, so these outputs cannot be scored as false positives, but a user sees them. This is why every report and every screen says that a class name is not a statement that the finding is present. It is a design limitation of the training data, not a threshold problem.

7.5.3 Quality of the Generated Reports

For the generated reports, quality was defined as faithfulness rather than style: the report must contain the same findings, values and review states as the evidence, in every language, and nothing else. Measured that way, the template generator met every check in the test campaign of Section 6.2 (61 of 61 checks, plus 21 API and 7 browser checks). The validator rejected each kind of invalid evidence that was tried, and a regenerated report contained identical findings. These results describe the generator's faithfulness to the model output; they say nothing about whether the model output itself is clinically correct.

7.6 VALIDATION OF RESULTS

7.6.1 Leakage Checks

Table 7.11 summarizes the leakage check of 2026-10-15: 17 of 18 checks passed and one gave a warning. No radiograph, identified by the hash of its file bytes, appears in two splits, and no image used to train a model or to select its epoch is in the test split. Near-duplicate candidates found by perceptual hashing were all rejected by a pixel-correlation test. The warning concerns three unlabeled *validation_data* images that duplicate test radiographs; they are not used in any metric. These checks rule out only identical or near-identical films across splits. DENTEX has no patient identifiers, so two different radiographs of the same patient could fall into different splits, and patient-level leakage cannot be excluded [39].

Table 7.11: Leakage Checks (Held-Out Test Split)

Check (unit: SHA-256 of file bytes)	Overlap	Result
All six split pairs (train, validation, calibration, test)	0	pass
Per task: training images vs. all test films (4 tasks)	0	pass
Per task: epoch-selection images vs. all test films	0	pass
Unlabeled pool (after duplicate exclusion) vs. test	0	pass
Approved-review training items vs. validation, calibration and test	0	pass
Near duplicates across splits (pixel correlation $r \geq 0.97$)	0	pass
validation_data (unlabeled) vs. test	3	warning

Source: Data/reports/leakage_check.md. Distinct annotated radiographs: train 794, validation 113, calibration 113, test 113.

7.6.2 Controlled Use of the Test Split

Use of the test split is gated in code. A test evaluation needs an explicit flag, a named authorizer, a written reason and a prior validation report for the same checkpoint hash; partial and repeated runs are refused, and every run is appended to a hash-chained ledger (Table 7.12). The four deployed checkpoints were evaluated on 2026-10-14, before the authorization fields existed, so their entries have no authorizer. Two further uses happened outside the ledger: a Kaggle re-run evaluated its BASE recipe once on the test split on 2026-10-16 (ground-truth-box macro-F1 0.693 ± 0.030 over five seeds), and an external-data run evaluated its five BASE seeds on the same split on 2026-10-24 (0.705 ± 0.031). Neither run selected or changed a deployed model, and neither result is used anywhere else in this report. Disclosure records were prepared for both, but the entries have not yet been added to the ledger.

Table 7.12: Recorded Uses of the Internal Test Split

#	Checkpoint or run	Task	Date	Films
1	dentex_fdi_numbering_v1	numbering	2026-10-14	63
2	dentex_findings_classification_v1	classification	2026-10-14	73
3	dentex_teeth_detection_v1	detection	2026-10-14	63
4	dentex_teeth_segmentation_v1	segmentation	2026-10-14	63
5	Classifier uncertainty evaluation	calibration	2026-10-14	crops
(6)	Kaggle BASE, seeds 42 to 46	classification	2026-10-16	73
(7)	External-data run, BASE seeds 42 to 46	classification	2026-10-24	73

Entries 1 to 5: Data/reports/test_set_ledger.json. Entries (6) and (7) are not yet in the ledger; sources: reports/p5_disclosure_entry.json and kaggle/artifacts_external_run1/ledger_disclosure_entry.json.

7.6.3 Reproducibility

Every evaluation report records the dataset fingerprint, the manifest and split hashes and the software versions, and the checkpoint hashes match those recorded at training time. Re-running the production trainer with seed 42 reproduced the deployed classifier bit for bit. Re-running the classifier's validation evaluation on the CPU gave accuracy 0.822 and macro-F1 0.711 against the stored 0.825 and 0.718, so results are of the same size across devices but not bit-identical. The four deployed checkpoints are legacy for provenance: they were trained before image-level training manifests existed. Their recorded image counts match the current split, which is consistent with, but does not prove, training on exactly these images.

7.6.4 Attempts to Improve the Classifier

Table 7.13 lists the controlled experiments run to improve the classifier. All used validation data only, five seeds and acceptance criteria fixed before the results were known: a detector-box macro-F1 gain above 0.027 in at least four of five paired seeds, no loss of Deep Caries or Periapical Lesion recall, and limits on per-class losses. No variant met them, and every alternative imbalance method lowered Deep Caries recall relative to the baseline. Five seeds of the production recipe gave detector-box macro-F1 between 0.651 (the deployed seed) and 0.776, so the deployed checkpoint is the weakest of five draws of its recipe; promoting the best seed would have been selection on validation, and no replacement was promoted.

Table 7.13: Classifier-Improvement Experiments (Validation, 5 Seeds)

Study	Variant	Macro-F1	Verdict
Class imbalance, GT boxes	inverse-frequency weights, cap 5 (C0)	0.764 ± 0.027	baseline
	weighted sampler	0.766 ± 0.009	not accepted
	oversampling	0.774 ± 0.016	not accepted
	focal loss ($\gamma = 2$)	0.750 ± 0.022	not accepted
	square-root weights	0.742 ± 0.024	not accepted
	no rebalancing	0.751 ± 0.017	reference
	minority geometric augmentation	0.751 ± 0.020	not accepted
Crop jitter, detector boxes	C0r (baseline)	0.723 ± 0.016	baseline
	jittered training boxes	0.720 ± 0.038	rejected
Training recipe, detector boxes	production recipe P	0.722 ± 0.045	baseline
	experimental recipe E	0.724 ± 0.015	rejected
Kaggle re-run, detector boxes	BASE	0.726 ± 0.026	baseline
	6 variants (best mean gain +0.004)	0.712 to 0.730	none accepted

Sources: docs/audit_evidence/ml_improvement/imbalance/report.md, Data/reports/crop_robustness_experiment.md, training_recipe_comparison_results.md, checkpoints/kaggle_candidates/dc9c505b/README.md. Ground-truth-box and detector-box rows are not comparable with each other.

7.6.5 Active Learning

An offline simulation measured whether uncertainty-based selection saves labels (Table 7.14). Labels already present in the training split were revealed on selection, and a ResNet-18 was retrained at each budget (160 pixels, 8 epochs, three seeds; pool of 476 radiographs; evaluation on the 331 validation crops). Neither entropy nor combined selection beat random selection: paired differences against random over budgets of 120

to 300 films averaged −0.005 for entropy (better in 7 of 12 comparisons) and −0.011 for combined selection (better in 4 of 12), well within seed variation [38]. The study measures label efficiency, not annotator effort.

Table 7.14: Active Learning: Validation Macro-F1 (Mean ± SD, 3 Seeds)

Labeled films	Random	Entropy	Combined
60	0.553 ± 0.053	0.553 ± 0.053	0.553 ± 0.053
120	0.626 ± 0.014	0.639 ± 0.011	0.618 ± 0.013
180	0.680 ± 0.019	0.665 ± 0.035	0.640 ± 0.013
240	0.712 ± 0.029	0.695 ± 0.010	0.702 ± 0.026
300	0.706 ± 0.014	0.706 ± 0.029	0.720 ± 0.027

Source: Data/reports/active_learning_experiment.md and .json; paired differences in docs/UNCERTAINTY_AND_ACTIVE_LEARNING.md.

7.6.6 Cross-Validation After the Main Evaluation

After the evaluation reported above, a radiograph-grouped five-fold cross-validation of the baseline ResNet-50 classifier recipe was run on the DENTEX training and validation pool (541 radiographs, 2,851 findings, five seeds). Its out-of-fold macro-F1 was 0.677 ± 0.008 on detector boxes and 0.698 ± 0.007 on ground-truth boxes (mean ± SD over seeds). The result is reported separately, was not used to select any model, and does not change the results above.

7.7 COMPARISON WITH EXISTING APPROACHES

A direct numerical comparison with published systems would be misleading. The studies of Chapter 2 use different datasets, splits, label sets and metrics, and the one shared benchmark, the official DENTEX test set, was excluded from this project because its label set could not be mapped to the four English classes and 95 of its 250 images duplicate training radiographs. Table 7.15 therefore compares evaluation settings rather than scores and states where a comparison is or is not possible.

Table 7.15: Comparison of Evaluation Settings with Existing Approaches

Approach	Usual evaluation in the literature	Evaluation in DentaMind AI	Directly comparable?
Tooth detection and numbering [11]–[13]	Per-tooth detection and numbering metrics on each study's own data	AP50, recall, exact and strict numbering accuracy on a 63-film internal test split	No: different data and splits
Finding detection and classification [17], [18]	Detection or diagnostic accuracy on each study's data	Classification of given crops, plus end-to-end on detector crops	No: different task definition
DENTEX baseline [15]	Official DENTEX test set	Official test set excluded (label mismatch, duplicates)	No
Image-to-report generation [20], [21]	Generated text against reference reports	Faithfulness checks of every value against the model evidence; no reference reports exist	No: different goal
Selective prediction [31]	Risk-coverage trade-off	Deployed entropy rule vs. the other rules among the 33 compared on the same crops (Table Table 7.9)	Yes, within this project
Ground-truth vs. detector crops	Usually only ground-truth crops	Both, on the same 331 findings (Table Table 7.10)	Yes, within this project

The comparisons that can be made fairly are internal: the same classifier on ground-truth and on detector crops, the deployed abstention rule against the other rules compared on the same crops, and the deployed training recipe against controlled variants. In all three, the deployed configuration was either the only one that met its criteria or no alternative was clearly better. Against the literature, the project's distinctive features are its end-to-end evaluation, its review routing with thresholds fixed on separate data, its evidence-checked multilingual reports and its documented data protocol, not higher accuracy.

7.8 SCREENSHOTS OF THE IMPLEMENTED SYSTEM

The following screenshots are genuine captures of the running application with the real models. They were taken during the project's own demonstration and audit sessions and are reproduced here in grayscale, as the report format requires black-and-white illustrations. The radiographs shown are DENTEX images or uploads used in those sessions. Captures from 2026-10-14 and 2026-10-15 predate the change of 2026-10-16, after which abstained regions also show their highest-scoring class with a review label.

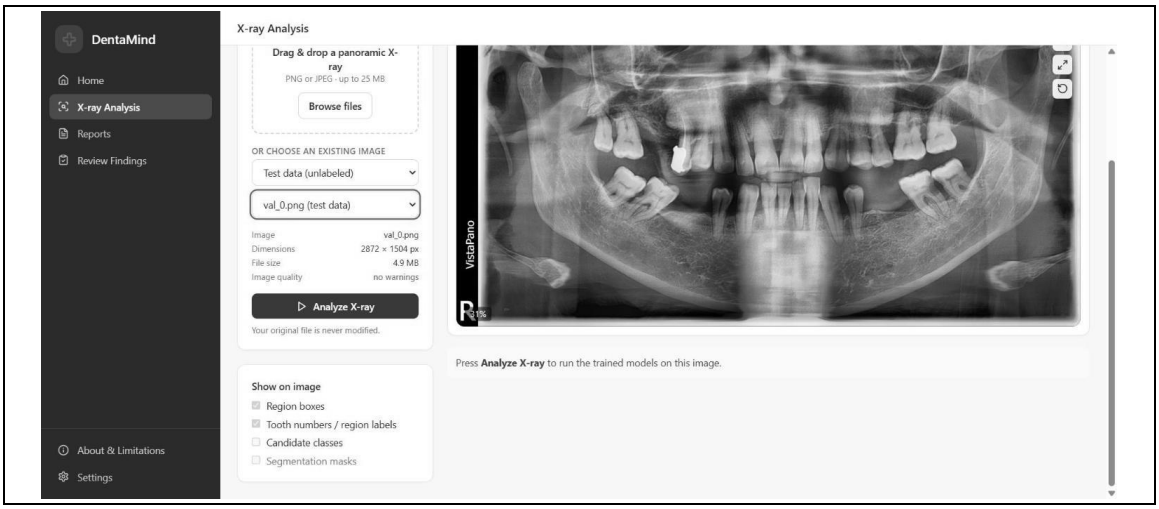


Figure 7.2: X-ray Analysis page with an unlabeled DENTEX image (val_0.png) selected before analysis (captured 2026-10-15).

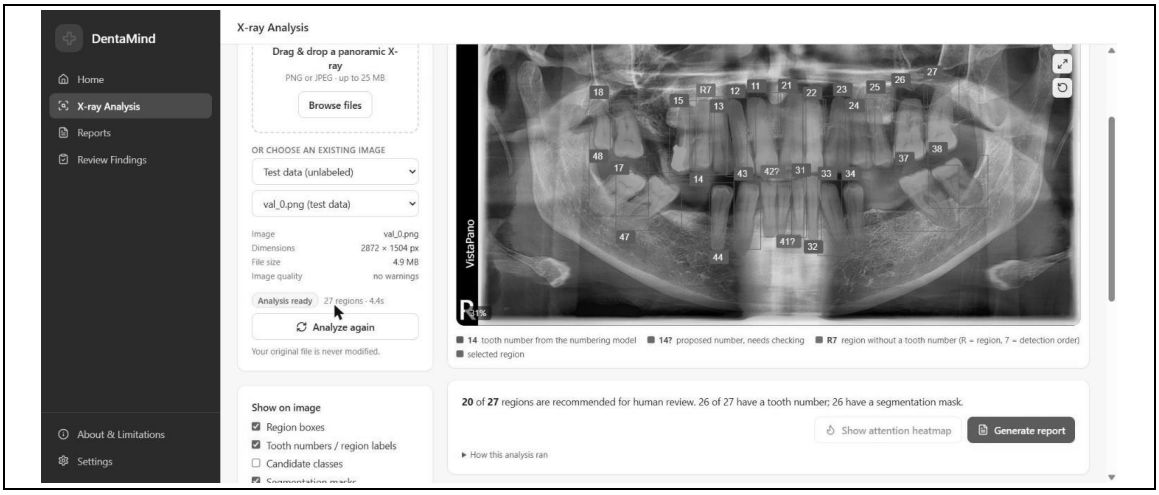


Figure 7.3: Analysis result for the same image: 27 regions, 26 with a tooth number and 26 with a mask; 20 regions recommended for human review (captured 2026-10-15).

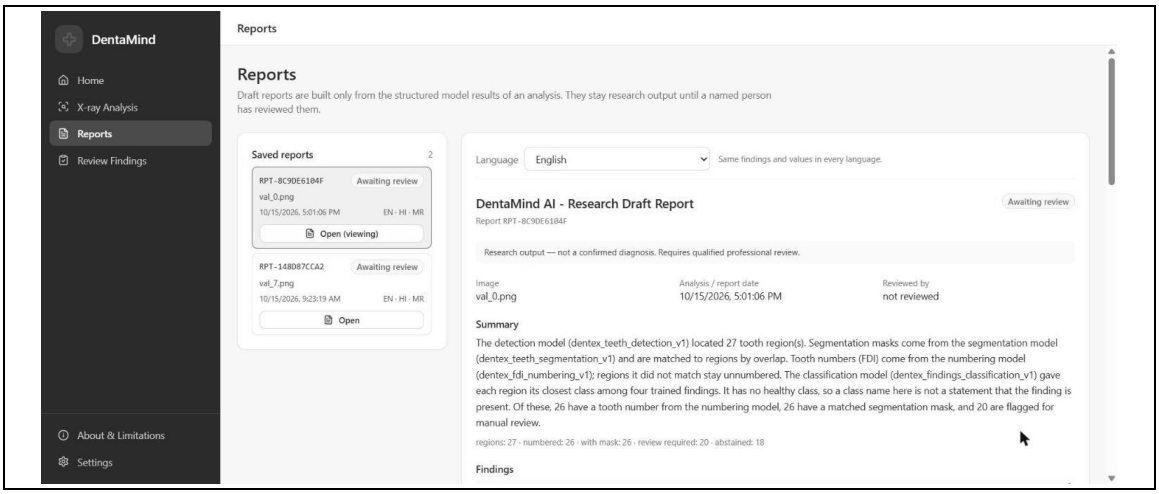


Figure 7.4: Reports page showing the English research draft generated for the same analysis, marked as awaiting review (captured 2026-10-15).

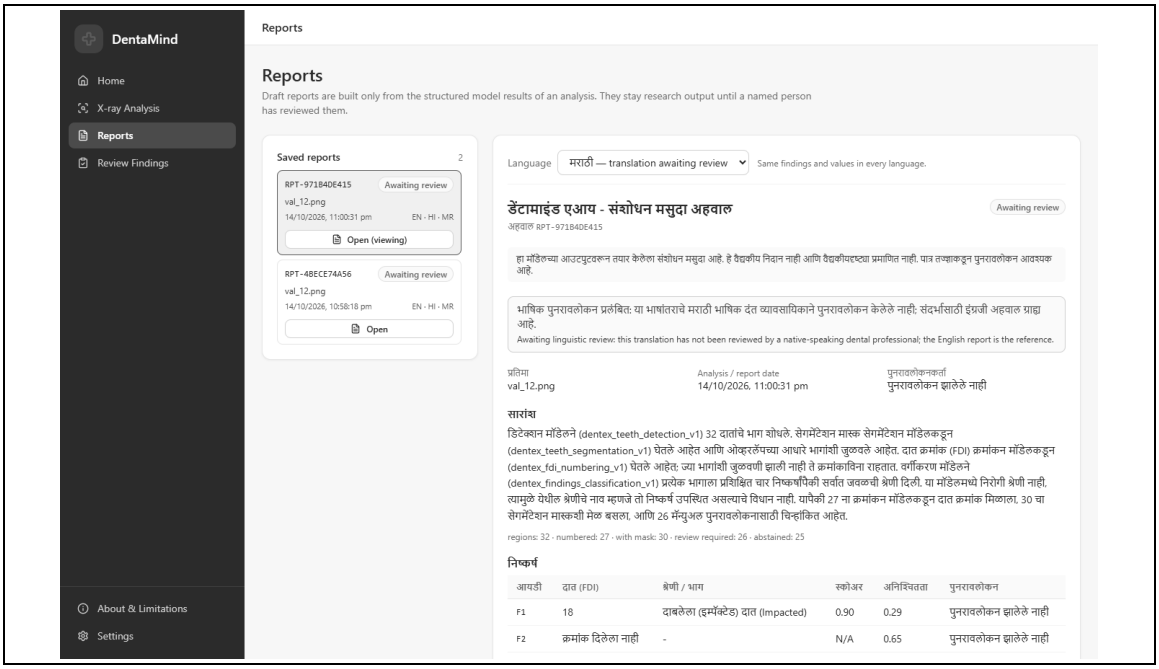


Figure 7.5: Marathi view of a generated report with the notice that the translation awaits linguistic review (audit capture, 2026-10-14).



Figure 7.6: Review Findings page: a region corrected by one reviewer, its revision history, and approval by a second named person (audit capture, 2026-10-14; test reviewer names).

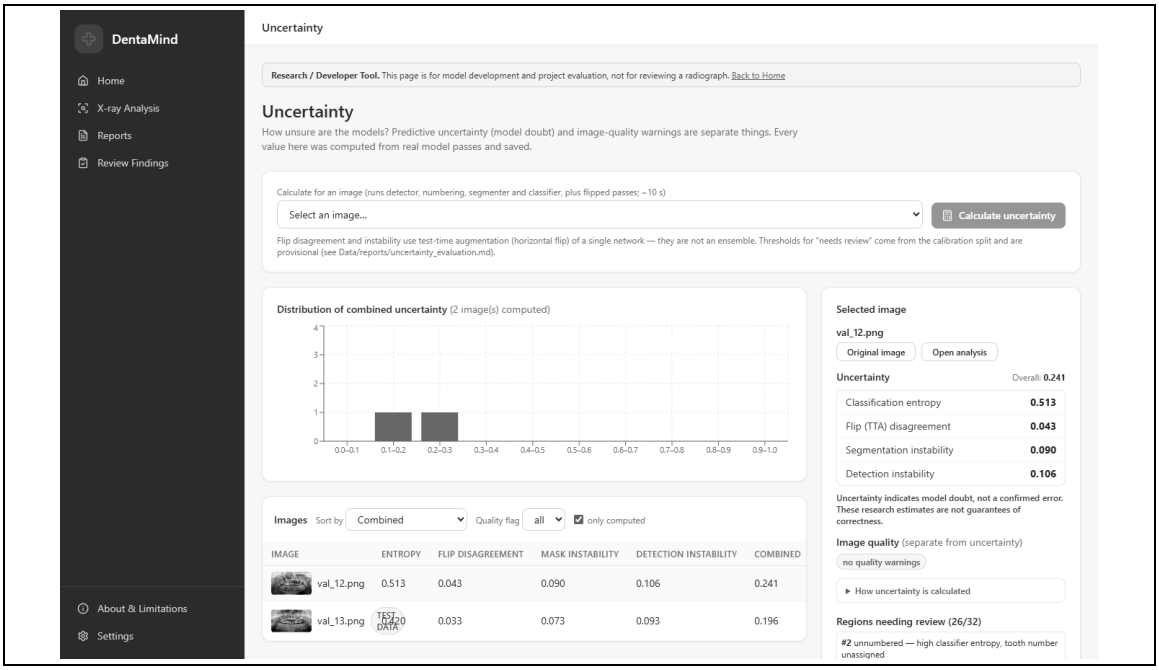


Figure 7.7: Uncertainty research page with per-image signals and the combined score (audit capture, 2026-10-14).

CHAPTER 8
CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

DentaMind AI shows that a complete panoramic-radiograph pipeline can be built from standard models, evaluated honestly on held-out data and wrapped in a tool that keeps a person in charge. On DENTEX, the localization models work well: test AP50 is 0.966 for tooth detection and mask AP50 is 0.974 for segmentation, and 96.8 % of matched teeth receive the exact FDI number. Finding classification is the weak link. Macro-F1 is 0.646 on ground-truth test crops, and on validation it drops from 0.718 to 0.634 once the classifier receives detector crops; Deep Caries and Periapical Lesion are recognized poorly in both settings.

The uncertainty signals carry real information about the classifier's errors, but with the current thresholds they capture most errors only by sending more than half of all regions to review. Four controlled attempts to improve the classifier and an active-learning study were all negative under criteria fixed in advance. These negative results are part of the contribution, because they mark the limits of what the system can claim.

The generative component did what it was designed to do. Every report is built only from the structured model output, all report-output checks passed, and the validator rejected every kind of invalid evidence that was tried. The design deliberately trades fluency for faithfulness. Whether a language model could make the reports more readable without breaking that faithfulness remains open, because the optional LLM path was never run against a live model.

8.2 LIMITATIONS OF THE PROPOSED SYSTEM

- **No clinical validation.** No study with qualified dentists has been carried out. The system is a research prototype, not a diagnostic device, and its reports are unconfirmed drafts.
- **No healthy class.** The classifier names the closest of four findings for every detected tooth, returns one label per tooth and produces uncalibrated scores.

- **Data limits.** One dataset from three institutions, one seed and one hold-out split; small validation and test splits (63 to 73 radiographs per task); only 9 validation and 13 test Periapical Lesions.
- **Possible patient-level leakage.** DENTEX has no patient identifiers, so different radiographs of one patient may fall into different splits.
- **Used test split.** The internal test split has been used, including two uses not yet recorded in the ledger, so an unbiased estimate needs a separate external test set.
- **Heavy review load.** The entropy flag sends more than half of the regions to review and still misses 13 % of errors; the thresholds are provisional.
- **Generative component untested live.** The LLM rewording path has only been tested with simulated replies, and the Hindi and Marathi wording has not been reviewed by native-speaking dental professionals.
- **Operational limits.** Reviewer names are self-declared; Label Studio, TensorBoard and Docker were not exercised; the four deployed checkpoints have no image-level training manifest.

8.3 FUTURE WORK

1. Evaluate the deployed models once on a separately sourced external test set under the documented protocol.
2. Obtain patient identifiers, or data that include them, so that splits can be grouped by patient.
3. Label teeth without findings so that a “no finding” class can be trained and real false-positive rates can be measured.
4. Move to a multi-label formulation for teeth that carry more than one finding.
5. Collect more Deep Caries and Periapical Lesion examples, the two classes that limit the classifier.
6. Carry out a reader study with qualified dentists to measure whether uncertainty-guided review improves accuracy or saves time.

7. Connect a language model to the LLM path, measure how often its output is rejected by the validator, and compare the readability of accepted drafts with the template drafts.
8. Have native-speaking dental professionals review the Hindi and Marathi templates.
9. Replace self-declared reviewer names with authenticated accounts before any shared use.

8.4 APPLICATIONS

Because the system is not clinically validated, its applications are in research and education rather than in patient care. The following uses are consistent with what has been measured:

- **Teaching aid.** Students can see tooth detection, FDI numbering and segmentation on real panoramic radiographs, together with the uncertainty of each prediction.
- **Annotation support.** The detector, numbering model and review queue can speed up the preparation of new labelled data, with every label confirmed by a person.
- **Research platform.** The end-to-end evaluation, leakage checks and test-set ledger can be reused to evaluate other dental models under a documented protocol.
- **Template for evidence-grounded reporting.** The evidence bundle, validator and localization check show one way to let a generator produce readable text without adding unsupported content, which applies beyond dentistry.

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APPENDIX A: PUBLICATION DETAILS

The template asks for the details of a paper publication, reviewer comments, certificates and the paper itself, or a draft paper if nothing has been published. The current status of each item is listed below. Nothing in this appendix should be read as evidence of acceptance or publication.

Table A.1: Status of Publication-Related Documents

Item	**Status (2026-10-26)**
Paper title	DentaMind AI: An Uncertainty-Aware Pipeline for Tooth Detection, FDI Numbering, Segmentation and Finding Classification on Panoramic Dental Radiographs
Authors	Nilesh Sonawane, Saurabh Mali, Sadhyoj Hanwate and Sachet Lothe (guide: Dr. Pradnya Narkhede)
Manuscript	IEEE two-column draft, 10 pages, 24 references (DentaMind_AI_IEEE_v5.pdf); reproduced below as the draft paper
Target conference or journal	PENDING: not yet chosen; the manuscript has not been submitted
Reviewer comments	NOT AVAILABLE: no submission has been made
Acceptance letter, publication or participation certificate	NOT AVAILABLE
Patent documents	NOT APPLICABLE: no patent application has been filed
Project event participation certificates	PENDING: to be attached by the team if any exist
Open items before submission	test-set ledger entries (6) and (7) need a named authorizer; the AI-assistance disclosure needs the authors' confirmation; an official similarity check has not been run

The draft manuscript follows, one page per report page.

DentaMind AI: An Uncertainty-Aware Pipeline for Tooth Detection, FDI Numbering, Segmentation and Finding Classification on Panoramic Dental Radiographs

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***Abstract*—DentaMind AI is a research prototype, built on the public DENTEX dataset, that detects teeth, assigns FDI numbers, segments tooth instances and gives each detected tooth the closest of four finding classes (Caries, Deep Caries, Impacted, Periapical Lesion), while routing uncertain regions to a human reviewer. On a held-out internal test split of 63 radiographs, the detector reached AP50 0.966 and the segmenter mask AP50 0.974 (mean matched Dice 0.873), and 96.8 % of matched teeth received the correct FDI number. On ground-truth crops from 73 test radiographs, the finding classifier reached accuracy 0.785 and macro-F1 0.646, but Periapical Lesion recall was only 0.308. With detector boxes in place of ground-truth boxes, validation macro-F1 fell from 0.718 to 0.634. An entropy rule flagged 54.2 % of test crops, which held 86.8 % of the classifier's errors; temperature scaling lowered test ECE from 0.048 to 0.036 but was not adopted. Four classifier-improvement attempts judged against criteria fixed in advance, and an active-learning study, were all negative. Splits are grouped by pixel hash and every use of the test split is logged, but DENTEX has no patient identifiers, so patient-level leakage cannot be ruled out. The system has not been clinically validated and is not a diagnostic device.**

***Index Terms*—Dental panoramic radiography, DENTEX, tooth detection, FDI tooth numbering, instance segmentation, finding classification, uncertainty estimation, calibration, selective prediction, data leakage, reproducibility.**

I. Introduction

A panoramic radiograph shows both dental arches in one exposure, which makes it a common first image in a dental examination and a demanding one to read: each tooth has to be located, numbered and checked for findings such as caries, impaction or periapical lesions. Deep learning has been applied to each of these steps, from tooth detection, numbering and segmentation [1], [2] to individual findings [3], [4], and the DENTEX benchmark lets enumeration and diagnosis be studied on the same public images [5], [6].

Two practical questions receive less attention. The first is how the stages behave together. Classifiers are usually scored on hand-drawn tooth boxes, while a deployed system crops whatever the detector returns, so errors carry over from one stage to the next. The second is whether the evaluation can be trusted. Leakage inflates reported performance across machine-learning research [7], and near-duplicate images shared between splits have been shown to raise medical-imaging benchmark scores [8]. On DENTEX, where one radiograph can appear in several sub-sets under different file names, this risk is concrete.

DentaMind AI was built to answer both questions on DENTEX. Its networks are standard (Faster R-CNN [9], Mask R-CNN [10] and ResNet-50 [11]), and we claim no architectural novelty. What we contribute is evidence about such a pipeline when every stage is measured on the same held-out data:

1) an open pipeline that links detection, FDI numbering, segmentation and finding classification region by region through IoU matching, never by list position (Figs. 1 and 2);
2) an end-to-end study of how detector boxes, instead of ground-truth boxes, change finding classification, with the loss traced to its causes;
3) an evaluation of uncertainty-based review routing, calibration and abstention, with thresholds fixed on a separate calibration split;
4) four controlled attempts to improve the classifier and one active-learning study, judged against criteria fixed before the results were known; all were negative; and
5) an auditable data protocol with pixel-hash grouped splits, leakage checks and a hash-chained ledger of every use of the test split.

Every number in this paper can be traced to a named evaluation report in the project repository. The system is a research prototype; it has not been clinically validated and does not make diagnoses.

II. Related Work

Detection, numbering and segmentation. Putra *et al.* [1] detect and number permanent teeth on panoramic radiographs with a deep detector. Zhang *et al.* [2] combine instance segmentation with FDI-compliant numbering in a multi-stage system that treats missing and supernumerary teeth explicitly. DentaMind AI covers the same task pair, but links the stages by IoU and adds review routing on top.

Finding classification. Szabó *et al.* [3] evaluate a commercial system for periapical lesions on 616 teeth from 357 panoramic radiographs; this is the class on which our classifier is weakest. Bittar *et al.* [4] summarize the evidence on AI for caries detection in an umbrella review. Hamamci *et al.* released DENTEX [5], the source of our data and finding taxonomy,

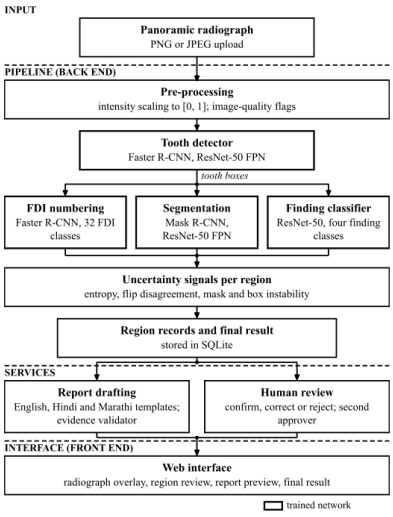


Fig. 1. System architecture of DentaMind AI as implemented. The tooth detector feeds number assignment, mask matching and finding classification; per-region uncertainty signals become review flags before the final result, report drafting and human review. Bold outlines mark trained networks.

together with a diffusion-based hierarchical detector for its enumeration-and-diagnosis task [6].

Uncertainty and data efficiency. From the uncertainty-quantification literature for medical images [12] we use only simple, single-network tools: selective classification, which lets a model withhold a prediction [13]; test-time augmentation as a cheap uncertainty estimate [14]; and temperature scaling [15], assessed with the expected calibration error (ECE) [16]. Active learning aims to cut labeling effort in medical imaging [17], and we test whether it does so here.

Data integrity. Kapoor and Narayanan [7] identify leakage as a common cause of irreproducible machine-learning results, and Huang *et al.* [8] show how near-duplicate contamination inflates a brain-MRI benchmark. Our split and audit protocol (Section III) is designed around both risks. Throughout, we measure how much existing techniques help on DENTEX rather than propose new ones.

III. Dataset and Data Preparation

A. DENTEX Sub-Sets and Their Overlap

We use the DENTEX panoramic radiographs [5], collected at three institutions with several scanners. Table I lists the sub-sets found by the project's dataset audit. The *quadrant-enumeration* sub-set labels every tooth with its quadrant and tooth index, which together give the two-digit FDI code [18]. The *quadrant-enumeration-disease* sub-set labels only teeth that carry one of four findings. Annotations include boxes and polygon masks.

TABLE I
DENTEX Sub-Sets Present in the Project (Dataset Audit, 2026-10-14)

Sub-set	Images	Annotations	Use
quadrant	693	2,772	not used by the current models
quadrant-enumeration	634	18,095	detection, numbering, segmentation
quadrant-enumeration-disease	705	3,529	finding classification
unlabelled	1,571	n/a	active-learning pool
validation_data (unlabeled)	50	n/a	inference demonstration only
All files on disk	3,653		2,716 distinct radiographs by pixels

Source: Data/reports/dataset_audit.md. In the disease sub-set, 27 images have no annotation.

The sub-sets overlap heavily. The 3,653 files hold only 2,716 distinct radiographs, and the 2,005 annotated entries only 1,133; 235 radiographs appear in all three annotated sub-sets, sometimes under different file names. Image counts quoted for DENTEX therefore count entries, not distinct radiographs, and certainly not independent patients.

B. Finding Taxonomy and Excluded Data

The disease sub-set contains 2,189 Caries, 578 Deep Caries, 604 Impacted and 158 Periapical Lesion annotations. It has no "healthy" label: a tooth without a finding is simply not annotated. The official DENTEX test release uses a different, Turkish and treatment-oriented label set (for example "1-çürük" and "3-kanal"), and the project found no verified equivalence between those labels and the four English classes. On the one radiograph shared by both releases, the English labels contradicted the mapping. In addition, 95 of the 250 official test images are byte-identical to radiographs in DENTEX's own training sub-sets. For both reasons, the official test set is excluded from every evaluation in this paper. The 50 unlabeled *validation_data* images are used only to demonstrate inference; 20 of them duplicate labeled radiographs.

C. Splitting Procedure

Each radiograph is identified by the SHA-256 hash of its pixels, so identical copies in different sub-sets form one group and always share a split. Groups are assigned with seed 42 to train, validation, calibration and test splits in proportions 0.70/0.10/0.10/0.10. The split file is append-only, so an image never changes split. Train fits weights, validation selects the epoch and gives validation metrics, calibration fixes thresholds and the temperature, and test is kept for final metrics. Table II gives the counts per task.

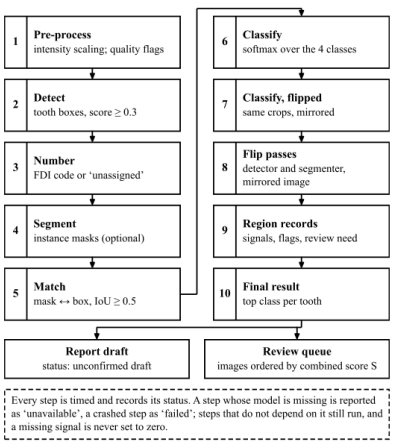


Fig. 2. End-to-end processing workflow: the ten implemented pipeline steps in execution order, the two outputs, and (dashed) the behavior when a step is unavailable or fails.

TABLE II
SPLIT STATISTICS PER TASK

Task (sub-set)	Train	Val.	Cal.	Test
Detection, numbering, segmentation: radiographs	440	69	62	63
Detection: tooth boxes	n/a	2,000	n/a	1,790
Findings: radiographs	476	65	64	73
Findings: Caries	1,550	205	206	228
Findings: Deep Caries	438	45	47	48
Findings: Impacted	412	72	55	65
Findings: Periapical Lesion	120	9	16	13

Sources: dataset_audit.md, eval_*.json and reports/training_summary.md. Finding counts are ground-truth finding boxes (crops); the calibration split has 324 finding crops. In training, Caries outnumber Periapical Lesion 12.9 : 1.

IV. System Architecture and Methods

A. Pipeline

Fig. 1 shows the implemented system and Fig. 2 its ten steps in execution order. A FastAPI back end runs the analysis, a SQLite store keeps images, analyses, reviews and reports, and a React front end displays them. The upload is never modified; models receive the RGB image as values in [0, 1], and torchvision's internal transform resizes it (shorter side 800 px, longer side at most 1,333 px); grayscale contrast statistics only raise image-quality flags. Each step is timed and records its status; if a model is missing or a step fails, independent later steps still run and the missing signal is reported as unavailable, never as zero. Region identifiers are derived from the image, the rounded box and the model version, so a review stays attached to the region the reviewer actually saw.

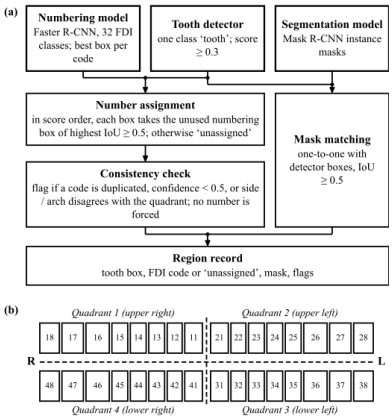


Fig. 3. (a) How detector boxes, the numbering model and the segmentation model are combined into a region record. (b) FDI two-digit tooth notation (ISO 3950) used for the 32 numbering classes, drawn as the teeth appear on a panoramic radiograph: the patient's right is on the image left, and molars are drawn wider.

B. Tooth Detection and Instance Segmentation

Tooth detection uses Faster R-CNN [9] with a ResNet-50 feature pyramid [11], [19] and a single class, "tooth"; detections with score ≥ 0.3 are kept. Instance segmentation uses Mask R-CNN [10] trained on the DENTEX polygon masks, so boxes are never converted into masks. Mask instances and detector boxes are linked by one-to-one greedy matching at intersection over union (IoU) ≥ 0.5, and mask quality is also summarized by the Dice coefficient:

$$\mathrm{IoU}(A,B) = \frac{|A \cap B|}{|A \cup B|}, \quad \mathrm{Dice}(A,B) = \frac{2|A \cap B|}{|A| + |B|} \tag{1}$$

where A and B are boxes or binary masks.

C. FDI Tooth Numbering

A second Faster R-CNN predicts 32 classes, one per permanent FDI code [18]. Because Faster R-CNN suppresses overlaps only within a class, one tooth can receive several numbers, so the raw output is resolved: boxes scoring below 0.3 are dropped, class-agnostic non-maximum suppression at IoU 0.5 is applied, and the best box per code is kept. Detector boxes are then visited in descending score order, and each takes the unused numbering box of highest IoU if that IoU is at least 0.5 (Fig. 3a). A tooth without such a box stays "unassigned"; the system never guesses a number. A rule-based check flags a number for review when it is duplicated, when its confidence is below 0.5, or when the box lies on the wrong side or arch for its quadrant (on a panoramic image, image-left is patient-right).

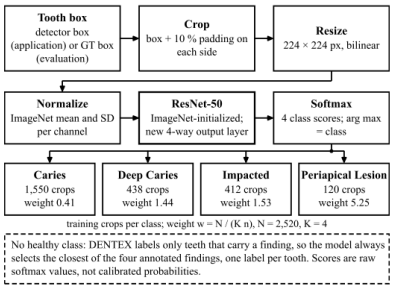


Fig. 4. Finding-classification workflow: tooth box, crop with 10 % padding, resize to 224 × 224, ImageNet normalization, ResNet-50 and a four-class softmax. Each class box gives its training-crop count and class weight from the checkpoint sidecar; the dashed note states that there is no healthy class.

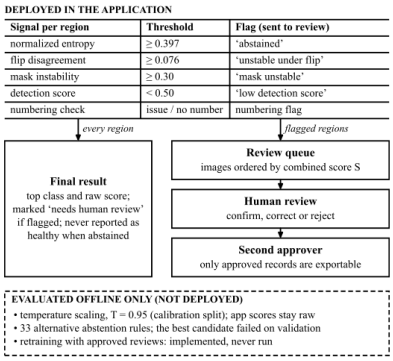


Fig. 5. Uncertainty-aware review routing. Upper part: thresholds and routing deployed in the application. Lower dashed panel: methods evaluated offline only, which are not active in the application.

D. Finding Classification

Each tooth box is cropped with 10 % padding, resized to 224 × 224 pixels, normalized with ImageNet statistics [20] and classified by a ResNet-50 with a new four-way output layer (Fig. 4). The application crops detector boxes, whereas the classifier evaluation crops ground-truth boxes; Section VI-D measures the difference. Because the training data contain no healthy class, the softmax answers "which of these four findings does this crop most resemble", not "is this tooth diseased". Training uses inverse-frequency class weights,

$$w_c = \frac{N}{K n_c} \tag{2}$$

where $N = 2{,}520$ training crops, $K = 4$ and n_c is the class count, which gives weights 0.41, 1.44, 1.53 and 5.25.

E. Uncertainty Signals

All signals come from a single network, not from an ensemble or a Bayesian model. For a softmax vector $\mathbf{p}$ over $K = 4$ classes, the normalized entropy is

$$H(p) = -\frac{\sum_k p_k \ln p_k}{\ln K} \tag{3}$$

Flip disagreement is the Jensen-Shannon divergence [21] between the prediction $\mathbf{p}$ on a crop and the prediction $\mathbf{q}$ on its horizontal flip, a two-pass form of test-time augmentation [14]:

$$\mathrm{JS}(p,q) = \frac{H'\left(\frac{p+q}{2}\right) - \frac{H'(p)+H'(q)}{2}}{\ln 2} \tag{4}$$

where H' is the unnormalized entropy. Mask instability and detection instability are both 1 − IoU between an instance and its best match on the flipped image (flipped back), and an instance without a counterpart counts as 1:

$$u_{\text{mask}} = 1 - \mathrm{IoU}\left(m, \tilde{m}_{\text{flip}}\right) \tag{5}$$

The image-level score that orders the review queue is the weighted mean of the available signals,

$$S = \frac{\sum_{j \in \text{avail}} \alpha_j s_j}{\sum_{j \in \text{avail}} \alpha_j} \tag{6}$$

with weights 0.4 (entropy), 0.4 (flip disagreement) and 0.2 (mask instability), renormalized over the signals that are available. Detection instability is displayed but excluded from S.

F. Review Routing, Human Review and Reports

Fig. 5 separates the thresholds deployed in the application from methods evaluated only offline. The entropy threshold 0.397 maximizes Youden's index J = TPR − FPR [22] on calibration crops, where a "positive" is a classifier error; the flip-disagreement threshold 0.076 is the 90th percentile on calibration crops. Both were fixed once, on one seed and one dataset, and are provisional. Algorithm 1 gives the routing as implemented. A region at or above the entropy threshold is "abstained" and sent for review. Since a project decision on 2026-10-16, its top class is still shown, marked as needing human review because of high classifier uncertainty, and an abstained tooth is never reported as healthy. Scores shown to users are raw softmax values, not calibrated probabilities.

A reviewer can confirm, correct or reject a region's class, set or clear its tooth number, or mark the image for specialist review or as unusable. Each record stores the model output the reviewer saw, a changed decision creates a new revision, and approval requires a second named person. Reviewer names are self-declared, and only approved records on unlabeled-pool or uploaded images can be exported for training. Reports are drafted from the structured findings with fixed templates in English, Hindi and Marathi. A validator rejects text that cites teeth, labels or numbers absent from the findings, and every report stays an unconfirmed research draft until a named reviewer signs it.

Algorithm 1: Per-image analysis and review routing (as implemented)

Input: radiograph I (RGB, values in [0, 1]); thresholds $\tau_{\mathrm{H}} = 0.397$, $\tau_{\mathrm{JS}} = 0.076$, $\tau_{\mathrm{M}} = 0.30$, $\tau_{\mathrm{D}} = 0.50$

$D \leftarrow$ tooth detector(I), keeping boxes with score ≥ 0.3
assign FDI codes to D from the numbering model (score order, IoU ≥ 0.5); run the consistency check
$M \leftarrow$ segmenter(I); match M to D one-to-one at IoU ≥ 0.5
repeat detection and segmentation on the flipped image (instability signals)
for each box $d \in D$ **do**
  $\mathbf{p} \leftarrow$ classifier(crop(d)); $\mathbf{q} \leftarrow$ classifier(flip(crop(d)))
  compute $H(\mathbf{p})$, JS($\mathbf{p}$, $\mathbf{q}$), u_{mask} and detection instability
  flags $\leftarrow$ {$H \geq \tau_{\mathrm{H}}$, JS $\geq \tau_{\mathrm{JS}}$, $u_{\mathrm{mask}} \geq \tau_{\mathrm{M}}$, score $< \tau_{\mathrm{D}}$, no matched mask, number needs review, not found on flipped image}
  status $\leftarrow$ "abstained" if $H \geq \tau_{\mathrm{H}}$ (top class still shown); review $\leftarrow$ flags $\neq \emptyset$ or number unassigned
$S \leftarrow$ weighted mean of the available image-level signals; store region records
draft the report as an unconfirmed draft; place the image in the review queue ordered by S

G. Training Configuration

Table III lists the configuration recorded in each checkpoint's sidecar file. All models are torchvision implementations. The detectors start from COCO weights [23] and the classifier from ImageNet weights; both are general-domain, not dental, starting points. Training uses AdamW with weight decay 10^{-4}, a cosine learning-rate schedule and seed 42, and keeps the epoch with the best validation score. Horizontal flips augment the one-class detector and the segmenter but not the numbering model, because a flip would swap left and right FDI codes. Training ran on an NVIDIA RTX 3050 Laptop GPU (4 GB) with PyTorch 2.6 and CUDA 12.4.

TABLE III
TRAINING CONFIGURATION OF THE FOUR DEPLOYED CHECKPOINTS

Model	Architecture (init.)	Epochs / batch / lr	Selection on val.
Tooth detector	Faster R-CNN R50-FPN, 1 class (COCO)	12 / 2 / 1e-4	box mAP@[.5:.95]
FDI numbering	Faster R-CNN R50-FPN, 32 classes (COCO)	16 / 2 / 1e-4	box mAP@[.5:.95]
Segmenter	Mask R-CNN R50-FPN, 1 class (COCO)	8 / 1 / 1e-4	box mAP@[.5:.95]
Finding classifier	ResNet-50, 4 classes (ImageNet V2)	15 / 32 / 3e-4	macro-F1 (epoch 5)

Sources: Data/checkpoints/*.json sidecars. Classifier augmentation: horizontal flip (p = 0.5) and multiplicative contrast jitter of ±5 %. The checkpoints are legacy: image-level training membership was not recorded (Section VIII).

V. EXPERIMENTAL SETUP AND EVALUATION METRICS

Fig. 6 summarizes the data protocol behind all splits, and Fig. 7 maps each task and study to its metrics and splits. Detection and segmentation use COCO-style average precision [23]: AP50 uses an IoU match threshold of 0.5, and mAP averages AP over IoU 0.50:0.05:0.95. At the operating point (score ≥ 0.3, IoU ≥ 0.5) we also report precision P, recall R and F1 from true positives (TP), false positives (FP) and false negatives (FN):

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}},\ R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}},\ \mathrm{F1} = \frac{2PR}{P+R} \tag{7}$$

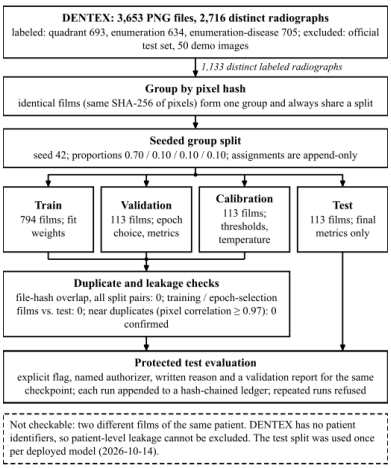


Fig. 6. Dataset splitting and integrity protocol: pixel-hash grouping, seeded split into four splits, duplicate and leakage checks, and the gated test evaluation. Film counts are distinct labeled radiographs; the dashed note states what cannot be checked.

Macro-F1 is the unweighted mean of the per-class F1 values. Per-class sensitivity, specificity and one-vs-rest ROC AUC are reported for the test split, and mask IoU is computed on a grid subsampled by a factor of four. Numbering is scored on ground-truth teeth: *coverage* is the fraction matched by a numbered prediction at IoU ≥ 0.5, *exact accuracy* is the fraction of matched teeth with the correct FDI code, and *strict accuracy* counts correct codes over all ground-truth teeth, so that missed teeth count as errors.

Calibration is measured by the expected calibration error over 15 equal-width confidence bins B_{m} [16] and by the multi-class Brier score [24],

$$\mathrm{ECE} = \sum_{m=1}^{15} \frac{|B_{\mathrm{m}}|}{n} |\mathrm{acc}(B_{\mathrm{m}}) - \mathrm{conf}(B_{\mathrm{m}})| \tag{8}$$

$$\mathrm{BS} = \frac{1}{n} \sum_{i} \sum_{k} (p_{ik} - y_{ik})^2 \tag{9}$$

where y_{ik} is the one-hot label. Temperature scaling [15] divides the logits $\mathbf{z}$ by a scalar T, fitted by negative log-likelihood on the calibration split over a grid from 0.5 to 5.0 in steps of 0.05:

$$p_k = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)} \tag{10}$$

For abstention rules [13], coverage is the fraction of crops that receive a class and risk is the error rate among them:

$$\mathrm{cov} = \frac{|\mathrm{acc.}|}{n},\ \mathrm{risk} = \frac{|\mathrm{wrong} \cap \mathrm{acc.}|}{|\mathrm{acc.}|} \tag{11}$$

How well a signal ranks errors is summarized by the AUROC (ties count one half).

Task	Metrics	Val.	Cal.	Test
MODELS				
Tooth detection	mAP, AP50, P / R @ 0.3	●		●
Segmentation	box / mask AP50, IoU, Dice	●		●
FDI numbering	AP50, coverage, accuracy	●		●
Classification, GT crops	P / R / F1, macro-F1, AUC	●		●
SYSTEM				
End-to-end, detector crops	matched / missed, macro-F1	●		
Calibration, uncertainty	ECE, Brier score, AUROC		●	●
Abstention rules	coverage, risk, macro-F1	●	●	
STUDIES				
Active learning	macro-F1 per label budget	●		
Classifier variants	paired macro-F1, 5 seeds	●		
Leakage check	overlaps, all split pairs	●	●	●
Inference benchmark	latency, peak memory	10 images		

Fig. 7. Experimental evaluation framework: each task or study, its metrics and the splits on which it was evaluated. Classification on ground-truth crops and end-to-end evaluation on detector crops are listed separately.

Three conditions apply to every result below. Validation also selected each model's training epoch, so validation numbers are optimistic. The internal test split was evaluated once for each of the four deployed checkpoints, on 2026-10-14, after model selection. All models were trained with one seed on one dataset, so no confidence intervals over seeds exist for the deployed models.

VI. Results

A. Tooth Detection and Segmentation

On the 63 test radiographs (1,790 teeth), the detector reaches AP50 0.966 and recall 0.974 at the operating point (Table IV). The segmenter's mask AP50 is 0.974, and matched instances have mean mask IoU 0.780 and mean Dice 0.873. The much lower mAP@[.5:.95] (0.515 for detection) means that boxes lie close to the annotations but are not tight, which matters once the classifier receives detector crops (Section VI-D).

TABLE IV
Tooth Detection and Instance Segmentation

Metric	Det. val	Det. test	Seg. val	Seg. test
Radiographs / GT teeth	69 / 2,000	63 / 1,790	69 / 2,000	63 / 1,790
Box mAP@[.5:.95]	0.560	0.515	0.569	0.530
Box AP50	0.978	0.966	0.981	0.974
Mask AP50	n/a	n/a	0.980	0.974
Mean mask IoU / Dice (matched)	n/a	n/a	n.r.	0.780 / 0.873
TP / FP / FN @ 0.3	1968/78/32	1743/73/47	1972/70/28	1754/69/36
Precision @ 0.3	0.962	0.960	0.966	0.962
Recall @ 0.3	0.984	0.974	0.986	0.980

Sources: eval_dentex_teeth_detection_v1_{val,test}.json and eval_dentex_teeth_segmentation_v1_{val,test}.json, rounded to three decimals. n.r. = not recorded in the validation report. Mask IoU on a 4× subsampled grid; 1,753 matched test instances.

B. FDI Tooth Numbering

On test, 96.9 % of ground-truth teeth are matched by a numbered prediction, and 96.8 % of those receive the exact FDI code (Table V). Counting missed teeth as errors gives a strict accuracy of 93.8 %. Quadrant accuracy is 0.999, so most errors put the wrong index within the right quadrant. The higher test than validation accuracy (0.968 vs. 0.936) is within small-sample variation for 63 and 69 radiographs, and we do not interpret it.

TABLE V
FDI Tooth Numbering

Metric	Validation	Test
Radiographs / GT teeth	69 / 2,000	63 / 1,790
Box mAP@[.5:.95] / AP50 (32 classes)	0.524 / 0.928	0.507 / 0.952
Precision / recall @ 0.3 (any class)	0.872 / 0.992	0.882 / 0.983
Coverage (matched / GT)	0.979 (1,957)	0.969 (1,734)
Exact accuracy among matched	0.936	0.968
Quadrant accuracy among matched	0.997	0.999
Strict accuracy over all GT teeth	0.916	0.938
Lowest per-tooth exact accuracy	36: 0.825; 47: 0.862	47: 0.912; 27: 0.917

Source: eval_dentex_fdi_numbering_v1_{val,test}.json. Third molars (18, 28, 38, 48) have the fewest ground-truth teeth (34 to 50 per split), so their per-tooth accuracies are the least stable.

C. Finding Classification on Ground-Truth Crops

On ground-truth test crops, the classifier reaches accuracy 0.785 and macro-F1 0.646 (Table VI). Impacted teeth are recognized reliably (F1 0.926, AUC 0.990) and Caries reasonably well (F1 0.842), but Deep Caries (F1 0.453) and Periapical Lesion (F1 0.364, recall 0.308) are poor. The confusion matrices in Table VII locate the main error: Caries and Deep Caries are confused in both directions (30 and 22 test crops), and 9 of the 13 test Periapical Lesions are labeled as some form of caries. The macro ROC AUC of 0.910 shows that the scores rank the classes better than the argmax decisions suggest.

TABLE VI
Finding Classification on Ground-Truth Crops

Class	Split	P	R	F1	n	Spec.	AUC
Caries	val	0.885	0.863	0.874	205		
	test	0.866	0.820	0.842	228	0.770	0.895
Deep Caries	val	0.500	0.467	0.483	45		
	test	0.414	0.500	0.453	48	0.889	0.834
Impacted	val	0.919	0.944	0.932	72		
	test	0.887	0.969	0.926	65	0.972	0.990
Periapical Lesion	val	0.467	0.778	0.583	9		
	test	0.444	0.308	0.364	13	0.985	0.921
Macro / accuracy	val	0.693	0.763	0.718	331	acc. 0.825	
	test	0.653	0.649	0.646	354	acc. 0.785	0.910

Source: eval_dentex_findings_classification_v1_{val,test}.json (65 and 73 radiographs). Specificity and one-vs-rest AUC were computed for the test split only. Validation macro-F1 was also the epoch-selection criterion.

TABLE VII
Confusion Matrices of the Finding Classifier (Rows: True Class)

True \ predicted	Caries	Deep C.	Impacted	Periap.
Caries (val / test)	177 / 187	18 / 30	5 / 6	5 / 5
Deep Caries	20 / 22	21 / 24	1 / 2	3 / 0
Impacted	2 / 1	2 / 1	68 / 63	0 / 0
Periapical Lesion	1 / 6	1 / 3	0 / 0	7 / 4

Each cell gives validation / test counts of ground-truth crops. Source: same reports as Table VI.

D. End-to-End Validation on Detector Crops

The classifier results above assume perfect boxes. A dedicated study ran the production path (detector, crop, classifier) on the 65 validation radiographs of the findings subset, with 331 findings, without touching calibration or test data (Table VIII). Production and evaluation preprocessing gave byte-identical pixels and identical predictions on all 331 crops, ruling out a preprocessing mismatch. On the 24 radiographs that also have a complete tooth annotation, the detector found 716 of 719 teeth at IoU 0.5 but only 582 at IoU 0.75. Detector boxes matched 323 of the 331 findings. Of the 8 unmatched findings, 4 were genuine detector misses; the other 4 were second findings on a tooth, since one detector box can match only one annotation.

TABLE VIII
END-TO-END VALIDATION (65 RADIOGRAPHS, 331 FINDINGS)

Measure	GT boxes	Detector boxes
Detection on 24 fully annotated films: P / R / F1 @ IoU 0.5	n/a	0.977 / 0.996 / 0.986
Same at IoU 0.75	n/a	0.794 / 0.809 / 0.802
Same 323 matched teeth: accuracy	0.830	0.811
Same 323 teeth: macro-F1	0.717	0.649
Same 323 teeth: macro recall	0.777	0.688
All 331 findings (miss = wrong): accuracy	0.825	0.792
All 331: macro-F1	0.718	0.634
All 331: Deep Caries recall (n = 45)	0.467	0.333
All 331: Periapical Lesion recall (n = 9)	0.778	0.444
Unlabeled detector boxes given a finding class	n/a	1,580 of 1,903

Source: Data/reports/end_to_end_validation_evaluation.md/.json (2026-10-15). Validation also selected the epochs, so all numbers are optimistic.

Accuracy fell from 0.825 to 0.792, a net loss of 11 findings: 20 findings became wrong with detector boxes, 14 became right, and 5 correct findings were lost to non-matching. About 10 % of predictions changed even when the detector box overlapped the annotation with IoU 0.8 to 1.0, so the classifier is sensitive to small crop shifts and not only to badly placed boxes. The loss falls mostly on the rare classes. Periapical Lesion recall fell from 7/9 to 4/9, and its precision on matched teeth from 0.46 to 0.29. Eight of the 331 findings (2.4 %), including 4 of the 9 validation Periapical Lesions, sit on teeth with a second finding, and a single-label classifier cannot be right for both.

The largest practical problem lies outside these metrics. Of the 1,903 detector boxes on the 65 radiographs, 1,580 match no annotated finding, yet without a "no finding" class the application names a closest class for each of them. On the 24 fully annotated films, 584 such boxes cover real teeth without a finding label, and the classifier called them Caries 475, Periapical Lesion 71, Deep Caries 23 and Impacted 15 times. DENTEX does not certify that unlabeled teeth are healthy, so these outputs cannot be scored as false positives, but the user sees them. This is a design limitation, not a threshold problem.

E. Attempts to Improve the Classifier

Table IX summarizes the controlled experiments run to improve the classifier. All used validation data, five seeds and acceptance criteria fixed in advance: a detector-box macro-F1 gain above 0.027 in at least four of five paired seeds, no loss of Deep Caries or Periapical Lesion recall, and limits on per-class losses. No variant met them, and every alternative imbalance method lowered Deep Caries recall relative to C0. Re-running the production trainer with seed 42 reproduced the deployed classifier bit for bit, while five seeds of the same recipe gave detector-box macro-F1 from 0.651 (the deployed seed) to 0.776. The deployed checkpoint is thus the weakest of five draws of its recipe; promoting the best seed would be selection on validation, so none was promoted. Crop jitter shifted the operating point toward the minority classes (Deep Caries recall +0.117) at the cost of precision instead of stabilizing predictions.

TABLE IX
CLASSIFIER-IMPROVEMENT EXPERIMENTS (VALIDATION, 5 SEEDS)

Study	Variant	Macro-F1	Verdict
Class imbalance, GT boxes	inverse-freq. weights, cap 5 (C0)	0.764 ± 0.027	baseline
	oversampling (best mean)	0.774 ± 0.016	not accepted
	sampler, focal loss (γ = 2), sqrt weights, minority aug., none	0.742 to 0.766	not accepted
Crop jitter, det. boxes	C0r (baseline)	0.723 ± 0.016	baseline
	jittered training boxes	0.720 ± 0.038	rejected
Training recipe, det. boxes	production recipe P	0.722 ± 0.045	baseline
	experimental recipe E	0.724 ± 0.015	rejected
Kaggle re-run, det. boxes	BASE	0.726 ± 0.026	baseline
	6 variants (best mean gain +0.004)	0.712 to 0.730	none accepted

Sources: docs/audit_evidence/ml_improvement/imbalance/report.md, Data/reports/crop_robustness_experiment.md, training_recipe_comparison_results.md, checkpoints/kaggle_candidates/dc9c505b/README.md. "None" is the no-rebalancing reference (0.751 ± 0.017). GT-box and detector-box rows are not comparable.

F. Inference Cost and Software Checks

On the RTX 3050 Laptop GPU, timed over 10 panoramic images of up to 3,076 × 1,536 pixels after two warm-up runs with no failures, mean inference took 0.110 s for detection (peak GPU memory 613 MB), 0.285 s for segmentation (1,653 MB), 0.334 s for detection plus numbering and 0.252 s for detection plus classification (the numbering harness runs the detector twice). In the archived UI audit of 2026-10-14, a full analysis of one image, flipped passes included, took 1.92 s. The project's 309 tests, which cover label mapping, split isolation, abstention rules, the final-result rule and resumed-training determinism, all passed after the last training run; they verify software behavior, not clinical performance.

VII. UNCERTAINTY, CALIBRATION AND HUMAN REVIEW

A. Error Capture and Calibration

Thresholds and the temperature were fitted on the calibration split and then applied once to the test crops (Table X). The entropy flag marks 54.2 % of test crops. Their error rate is 34.4 %, against 6.2 % for unflagged crops, and they contain 86.8 % of all errors (66 of 76). The flag therefore catches most errors, but it still misses 13.2 % of them and sends more than

half of all regions to review. The signals do carry information about errors (AUROC 0.768 for entropy and 0.745 for flip disagreement), although the simpler 1 − max-probability score ranked test errors slightly better (0.790).

TABLE X
UNCERTAINTY AND CALIBRATION OF THE FINDING CLASSIFIER

Measure	Calibration	Test
Crops / accuracy	324 / 0.830	354 / 0.785
ECE (15 bins) / Brier, raw scores	0.071 / 0.277	0.048 / 0.291
ECE / Brier after temperature T = 0.95	0.076 / 0.276	0.036 / 0.291
AUROC for errors: entropy	0.721	0.768
AUROC: flip disagreement	0.709	0.745
AUROC: 1 − max probability	0.713	0.790
Flagged by entropy ≥ 0.397	threshold fitted	54.2 %
Error rate flagged / unflagged	n/a	0.344 / 0.062
Errors inside flagged set	n/a	0.868
Flip-unstable (≥ 0.076): share; error rate vs. stable	90th pct.	9.3 %; 0.485 vs. 0.187

Source: Data/reports/uncertainty_evaluation.json (2026-10-14; test-ledger entry 5). Temperature scaling is not applied in the application.

Temperature scaling gave a mixed result. The fitted temperature (0.95) lowered test ECE from 0.048 to 0.036 but raised calibration ECE from 0.071 to 0.076 on the very split it was fitted on, and it left the Brier score almost unchanged. Because the effect is small and inconsistent between splits, the application keeps raw scores and labels them as uncalibrated.

B. Flip Instability of the Detector and Segmenter

A check on 30 validation radiographs, run through the live API, asked whether flip instability flags poor masks or false detections. Mask instability ranked masks with IoU below 0.8 (51.8 % of 851 masks) only weakly, with AUROC 0.593; masks above the median signal had an error rate of 0.598 against 0.439 at or below it. Detection instability ranked the 27 false positives among 878 detections (3.1 %) with AUROC 0.644, an estimate that rests on few events. Neither is a validated error detector (source: instability_validation.json).

C. Abstention Rules

An offline experiment compared 33 abstention rules (entropy, maximum-probability and top-2-margin thresholds) on detector crops (Table XI). Rules were selected on calibration and confirmed on validation, and the test split was never read. Only "abstain if max $p < 0.75$" passed the calibration criteria. On validation it raised coverage from 0.520 to 0.638 at almost the same risk (bootstrap 95 % CI of the risk difference −0.018 to +0.026), but it failed the minority-class criterion: Periapical Lesion precision fell from 0.50 to 0.43. The production entropy rule was therefore kept. The candidate would also have named a class for 345 instead of 250 of the 584 unlabeled tooth boxes, with unmeasurable correctness.

TABLE XI
ABSTENTION RULES ON DETECTOR CROPS

Rule / split	Coverage	Risk	Macro-F1 (all)	PL P / R
Entropy ≥ 0.397, cal.	0.506	0.093	0.570	0.80 / 0.50
max p < 0.75, cal.	0.623	0.096	0.620	0.75 / 0.56
Entropy ≥ 0.397, val.	0.520	0.119	0.518	0.50 / 0.43
max p < 0.75, val.	0.638	0.121	0.571	0.43 / 0.43

Source: docs/audit_evidence/ml_improvement/abstention/report.md (2026-10-15). Calibration 318 and validation 323 matched findings. "All" counts abstained and missing crops as not recalled. PL = Periapical Lesion; precision is over accepted crops.

D. Active Learning and Human Review in Practice

An offline simulation measured whether ranking unlabeled pool radiographs by uncertainty saves labels [17]. Labels already present in the training split were revealed on selection, and a ResNet-18 was retrained at each budget (160 px, 8 epochs, three seeds; pool of 476 radiographs; evaluation on the 331 validation crops). Neither entropy nor combined selection beat random selection. Paired differences against random over budgets of 120 to 300 films averaged −0.005 for entropy (better in 7 of 12 comparisons) and −0.011 for combined selection (better in 4 of 12), well within seed variation. At 300 labeled films, validation macro-F1 was 0.706 ± 0.014 (random), 0.706 ± 0.029 (entropy) and 0.720 ± 0.027 (combined). The study (source: active_learning_experiment.md) measures label efficiency, not annotator effort, and the Label Studio loop was not run against a live server.

The review mechanism is implemented and unit-tested, but its clinical value is unmeasured. No study with qualified dental reviewers has been run, reviewer identities are not verified, and retraining with approved reviews is implemented but has never been executed. In the archived UI audit on an unlabeled DENTEX validation radiograph, the application reported 32 regions, 27 with a number and 30 with a mask, and recommended 26 of the 32 for review. With the present thresholds, much of the work moves to the reviewer, as the flag rate in Table X predicts.

VIII. DATA INTEGRITY AND REPRODUCIBILITY

The leakage check run on 2026-10-15 passed 17 of 18 checks and gave one warning (source: leakage_check.md). By SHA-256 of file bytes, no radiograph appears in two splits, no image used for training or epoch selection is in the test split, and neither the deduplicated unlabeled pool nor approved-review items overlap the held-out splits. Near-duplicate candidates found by perceptual hashing were all rejected by a pixel-correlation test ($r \geq 0.97$). The warning concerns three unlabeled *validation_data* images that duplicate test radiographs; they are used in no metric. The splits contain 794, 113, 113 and 113 distinct annotated radiographs (train, validation, calibration, test). These checks rule out only identical or near-identical films. DENTEX has no patient identifiers, so two radiographs of one patient could fall into different splits, and patient-level leakage cannot be excluded [7].

Use of the test split is gated in code. A test evaluation needs an explicit flag, a named authorizer, a written reason and a prior

validation report for the same checkpoint hash. Partial and repeated runs are refused, and every run is appended to a hash-chained ledger (Table XII). The four deployed checkpoints were test-evaluated on 2026-10-14, before the authorization fields existed, so their entries have no authorizer. We also found two uses outside the ledger. The Kaggle re-run evaluated its BASE recipe once on the test split on 2026-10-16 (ground-truth-box macro-F1 0.693 ± 0.030 over five seeds), and a later external-data run evaluated its five BASE seeds on the same split on 2026-10-24 (0.705 ± 0.031). Neither run selected or changed a deployed model, and neither result is used elsewhere in this paper. Disclosure records were prepared for both, but the entries have not yet been appended to the ledger.

TABLE XII
RECORDED USES OF THE INTERNAL TEST SPLIT

#	Checkpoint / run	Task	Date	Films
1	dentex_fdi_numbering_v1	numbering	2026-10-14	63
2	dentex_findings_classification_v1	classification	2026-10-14	73
3	dentex_teeth_detection_v1	detection	2026-10-14	63
4	dentex_teeth_segmentation_v1	segmentation	2026-10-14	63
5	classifier uncertainty evaluation	calibration	2026-10-14	crops
(6)	Kaggle BASE, seeds 42 to 46	classification	2026-10-16	73
(7)	External-data run, BASE seeds 42 to 46	classification	2026-10-24	73

Entries 1 to 5: Data/reports/test_set_ledger.json. Entries (6) and (7) are not in the ledger; sources: reports/p5_disclosure_entry.json and kaggle/artifacts_external_run1/ledger_disclosure_entry.json.

Every evaluation report records the dataset fingerprint, manifest and split hashes and software versions, and checkpoint SHA-256 values match those recorded at training. The four deployed checkpoints, however, predate image-level training manifests; their recorded image counts match the current split, which is consistent with, but does not prove, training on exactly these images. Later checkpoints carry a manifest that the model registry verifies.

IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

All models were trained and tested on one dataset from three institutions, with one seed and a single hold-out split, so performance on other scanners, pediatric patients or unusual images is unknown. The validation and test splits are small (63 to 73 radiographs per task), and Periapical Lesion has only 9 validation and 13 test findings, so its metrics move in large steps when a single crop changes. Validation numbers are optimistic because validation chose the epochs, and since the internal test split has now been used, an unbiased estimate needs an external test set.

The classifier has no healthy class, returns one label per tooth, names a closest finding for every detected tooth, and produces uncalibrated scores. The entropy flag sends more than half of the regions to review and still misses 13 % of errors. The thresholds are provisional, and no study has measured whether review by qualified clinicians improves accuracy or safety. Reviewer identities are self-declared, and the Hindi and Marathi report wording has not been reviewed by native-speaking dental professionals.

The deployed application applies only the routing of Fig. 5 and Algorithm 1 to raw scores. Temperature scaling, the alternative abstention rules, retraining with approved reviews and the active-learning simulation were evaluated offline and are not active. DentaMind AI does not diagnose; its reports are unconfirmed drafts that require review by a qualified professional, and we make no claim of clinical validation, autonomous diagnosis or improved patient outcomes. DENTEX images are de-identified public research data released under the CC BY-NC-SA 4.0 license [5], whose terms apply to any reuse. All figures are schematic diagrams, and no radiograph is reproduced.

X. CONCLUSION

On DENTEX, standard detectors localize and number teeth well: test AP50 is 0.966 for detection and mask AP50 0.974 for segmentation, and 96.8 % of matched teeth receive the exact FDI number. Finding classification is the weak link. Macro-F1 is 0.646 on ground-truth test crops; on validation it drops from 0.718 to 0.634 once the classifier receives detector boxes, and Deep Caries and Periapical Lesion are recognized poorly in both settings. Uncertainty flags capture most classifier errors only by routing over half of all regions to review, no improvement attempt met its pre-set criteria, and active learning did not beat random selection. These negative results mark the limits of what the system can claim and point to what is needed next: more rare-class labels, labels for teeth without findings (enabling a “no finding” class and real false-positive rates), a multi-label formulation, patient-grouped splits, an external test set evaluated once, and a reader study with qualified dentists.

After the evaluation reported here, a radiograph-grouped five-fold cross-validation of the baseline ResNet-50 classifier recipe was run on the DENTEX training and validation pool (541 radiographs, 2,851 findings, five seeds). Its out-of-fold macro-F1 was 0.677 ± 0.008 on detector boxes and 0.698 ± 0.007 on ground-truth boxes (mean ± SD over seeds). It was not used to select any model and does not change the results above.

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APPENDIX B: PLAGIARISM REPORT

PENDING. The official plagiarism (similarity) report is not attached because it has not been generated. The PCCOE instructions make this report mandatory: the similarity must be below 15 %, it must be produced with a good-quality similarity-checking tool, and it must be verified by the guide.

No similarity score is stated anywhere in this report, and none should be inferred. The team must run the official check on the final text of this report, have the guide verify it, and insert the report here before submission.